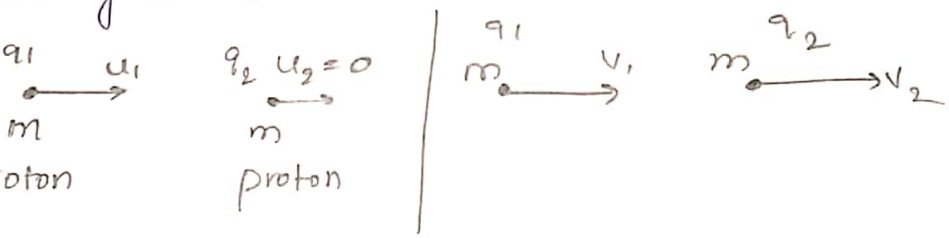
Collisions

Collision: A physical attraction b/w two or more bodies that exert strong forces each other for a short duration and cause finite change in momentum for each colliding body

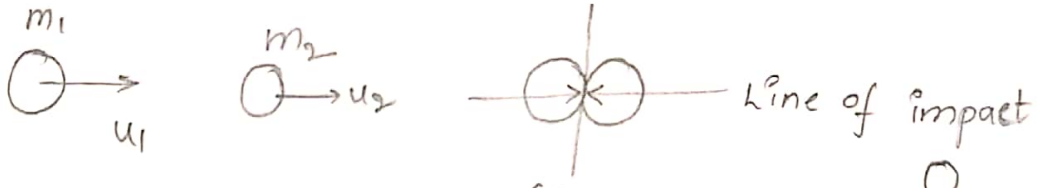
* during collision, bodies may or may not be in physical contact

Ex: ① Two billiard balls / vehicles are in contact during collision.

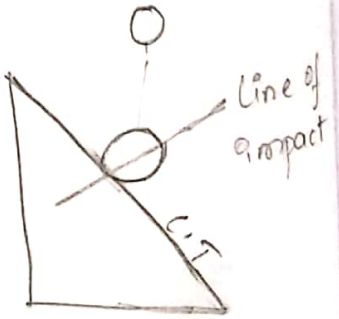
② 

electrostatic forces are the interaction forces.

Line of Impact:



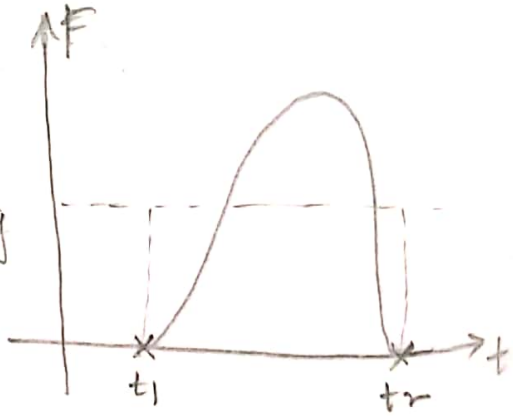
* Along the line of impact bodies exert strong forces on each other.



* Change in momentum of each body occurs along line of impact.

Impulse:

we have, $J = \int_{t_1}^{t_2} F\,dt$ → impulsive force.



$\bar{J}$ = Area of F-t graph b/w $t = t_1$ & $t = t_2$.

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$$\boxed{\bar{J} = f_{avg} \cdot \Delta t} \quad \text{---} (1)$$

Again, $\bar{F} = \frac{d\bar{p}}{dt}$

$$d\bar{p} = \bar{F}\, dt$$

$$J = \int_{P_1}^{P_2} d\bar{p} = \bar{P}_2 - \bar{P}_1 = \Delta\bar{P} \quad \text{---} (2)$$

from eq (1) & (2),

$$\Delta\bar{P} = \bar{F}_{avg}\, \Delta t$$

$$\bar{F}_{avg} = \frac{\Delta\bar{P}}{\Delta t} = \frac{\bar{P}_f - \bar{P}_i}{t_f - t_i}$$

∴ In general,

$$\boxed{\bar{P}_i \pm \bar{J} = \bar{P}_f} \quad \text{for)} \quad \bar{P}_i \pm \int_{t_1}^{t_2} \bar{F}\, dt = \bar{P}_f$$

Examples for impulsive forces:

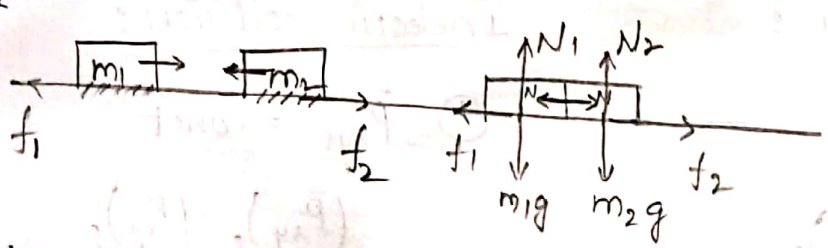
1) Normal reaction, tension, friction are impulsive forces which are case dependent.

2) impulsive force is a relative term.

3) gravitational force, spring force are always non-impulsive

→

*

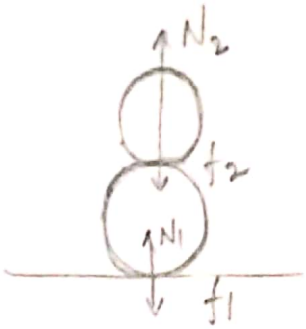


Here, N_1, N_2 are non-impulsives.

⇒ f_1, f_2 are non impulsive.

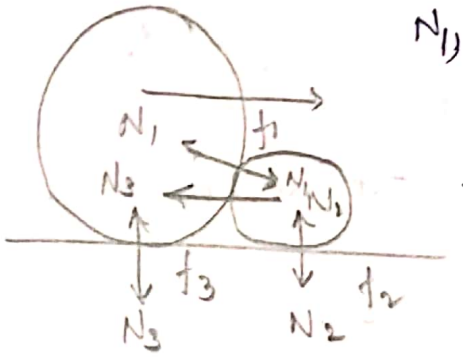
but 'N' is impulsive force.

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Both N_1, N_2 are impulsive forces

$\Rightarrow$ f_1, f_2 are also impulsive forces.



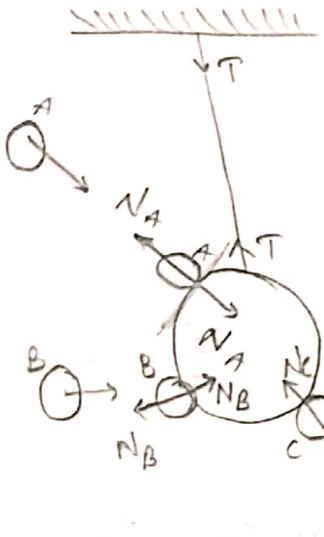
N_1, N_2 are impulsive forces.

N_3 — Non-impulsive.

f_1, f_2 — Impulsive

f_3 — non-impulsive.

*



*Tension — impulsive. (only in case of collision of A).

$N_A \longrightarrow$ impulsive

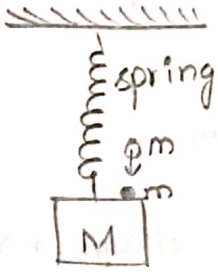
* $N_B \longrightarrow$ impulsive

but T is non-impulsive.

* N_C — impulsive

but T is non-impulsive.

*



there will be no sudden changes in spring bcoz it has more inertia.

∴ spring force is non-impulsive.

Elastic collision :

① P_{system} = const

$$(\bar{P}_{sy})_i = (\bar{P}_{sy})_f$$

② K.E of system is constant

Inelastic collision :

① $\bar{P}_{sys}$ = const

$$(\bar{P}_{sy})_i = (\bar{P}_{sy})_f$$

② K.E of system is not constant.

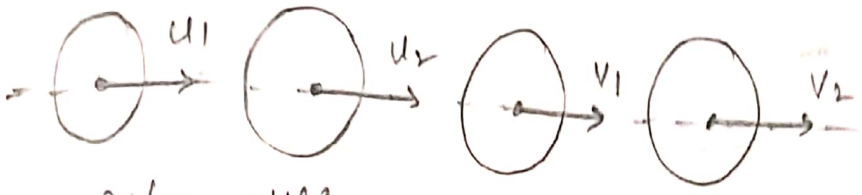
③ Deformation P.E = $KE_i - KE_f$

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Collisions

- Head on collision [1D collision]
- [2D - collision]

1D:



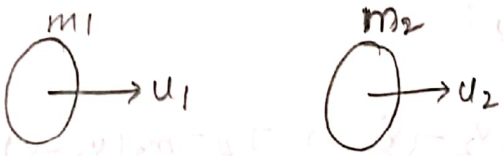
Before collision — After collision

2D:



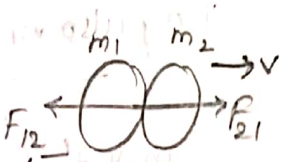
Before collision — After collision

Coefficient of Restitution (e):



$(u_1 > u_2)$

Just before collision



$(\vec{F}_{12} = -\vec{F}_{21})$

During collision [Max deformation]

* Impulse on m_1,

$$m_1u_1 - \int F_{12}\cdot dt = m_1v$$

$\int F_{12}\cdot dt$ = Impulse of deformation (J_D)

$$m_1u_1 - J_D = m_1v \quad \text{— (1)}$$

$$\boxed{J_D = m_1(u_1 - v)}$$

* Impulse on m_2,

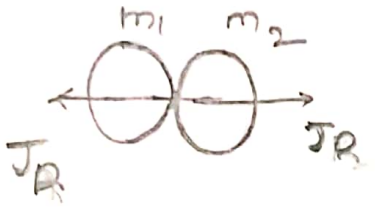
$$m_2u_2 + J_D = m_2v \quad \text{— (2)}$$

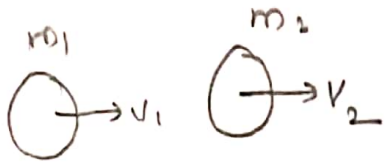
$$\boxed{J_D = m_2(v - u_2)}$$

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$$m_1u_1 + m_2u_2 = (m_1+m_2)V$$

$$P_i = P_f \Rightarrow \boxed{V = \frac{m_1u_1 + m_2u_2}{m_1+m_2}} \quad \text{—(a)}$$





Just after collision
$(v_2 > v_1)$

[During reformation]

* During reformation,

Impulse on m_1:

$$m_1v - J_R = m_1v_1 \Rightarrow \boxed{J_R = m_1(v-v_1)} \quad \text{—(1)}$$

Impulse on m_2:

$$m_2v + J_R = m_2v_2 \text{ —(2)} \Rightarrow J_R = m_2(v_2 - v)$$

Again, adding (1) & (2)

$$m_1v_1 + m_2v = m_1v_1 + m_2v_2$$

$$\boxed{v = \frac{m_1v_1 + m_2v_2}{m_1+m_2}} \quad \text{—(b)}$$

from eqn (a) & (b)

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

now, coefficient of restitution,

$$\boxed{e = \frac{J_R}{J_B}}$$ for any colliding body.

$$1/e = \left(\frac{m_1(u_1-v)}{m_1(v-v_1)}\right) \Rightarrow e = u_1 - \left(\frac{m_1u_1 + m_2u_2}{m_1+m_2}\right)$$

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$$e = \frac{u_1 - \frac{m_1u_1 + m_2u_2}{m_1+m_2}}{\frac{m_1v_1 + m_2v_2}{m_1+m_2} - v_1}$$

$$= \frac{m_1u_1 + m_2u_2 - m_1u_1 - m_2u_2}{m_1v_1 + m_2v_2 - m_1v_1 - m_2v_1}$$

$$= \frac{m_2(u_1 - u_2)}{m_2(v_2 - v_1)}$$

$$\boxed{e = \frac{u_1 - u_2}{v_2 - v_1}} \Rightarrow \boxed{e = \frac{v_2 - v_1}{u_1 - u_2}}$$

Points:

* e has no units & dimensions.
* e is purely a number
* $\boxed{0 \le e \le 1}$
* If e=1 then it is elastic collision.
* e=0 then it is perfectly in-elastic collision.
* $e = \frac{v_2 - v_1}{u_1 - u_2} = 0$

$$v_2 - v_1 = 0$$

$$\boxed{v_1 = v_2}$$

i.e, both the bodies move together just after collision in perfectly inelastic collision.

* In exact form,

$$e = \frac{|\bar{v}_2 - \bar{v}_1|}{|\bar{u}_1 - \bar{u}_2|} = \frac{\text{relative velocity of separation of bodies after collision}}{\text{relative velocity of approach of bodies before collision}}$$

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* $\bar{u}_1, \bar{u}_2 \rightarrow$ velocity components of bodies along line of Impact before collision.

$\bar{v}_1, \bar{v}_2 \rightarrow$ velocity components of bodies along line of impact after collision.

* e doesnot depends on size, shape, masses and velocities of bodies.

* It depends only on nature of material of colliding bodies.

Maximum deformed P.E stored by the system:



(maximum deformed state)

* $U_{max} = K.E_i - K.E_f$

$$= \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 - \frac{1}{2}(m_1+m_2)v^2$$

$$= \frac{1}{2}m_1(u_1^2 - v^2) + \frac{1}{2}m_2(u_2^2 - v^2)$$

$$= \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 - \frac{1}{2}(m_1+m_2)\left(\frac{m_1u_1+m_2u_2}{m_1+m_2}\right)^2$$

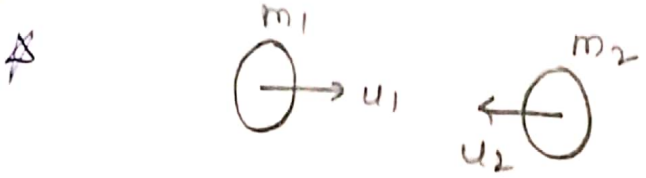
$$= \frac{1}{2}\left[m_1u_1^2 + m_2u_2^2 - \frac{(m_1u_1+m_2u_2)^2}{m_1+m_2}\right]$$

$$= \frac{1}{2}\left[\frac{m_1^2u_1^2 + m_1m_2u_1^2 + m_1m_2u_2^2 + m_2^2u_2^2 - m_1^2u_1^2 - m_2^2u_2^2 - 2m_1m_2u_1u_2}{m_1+m_2}\right]$$

$$\boxed{U_{max} = \frac{1}{2}\left[\frac{m_1m_2}{m_1+m_2}(u_1-u_2)^2\right]}$$

* In general, $U_{max} = \frac{1}{2}\frac{m_1m_2}{m_1+m_2}(\bar{u}_1 - \bar{u}_2)^2$

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$$U_{max} = \frac{1}{2}\frac{m_1 m_2}{m_1+m_2}\left(u_1 - u_2(-1)\right)^2$$

$$= \frac{1}{2}\frac{m_1 m_2}{m_1+m_2}(u_1+u_2)^2$$

Points:

① For any collision,

L.C.L.M is valid at all times

$$\boxed{m_1\vec{u_1} + m_2\vec{u_2} = m_1\vec{v_1} + m_2\vec{v_2}}$$

② Even in elastic collision,

K.E of system is not conserved during collision.

∴ Mechanical energy of system is also not conserved during collision.

But Total Energy of system is always conserved at all times.

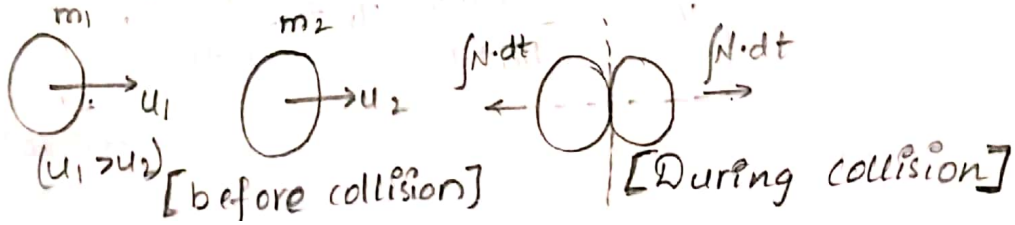
③ In elastic collisions (just after collision is completed)

$$\boxed{K.E_{system} = const}$$

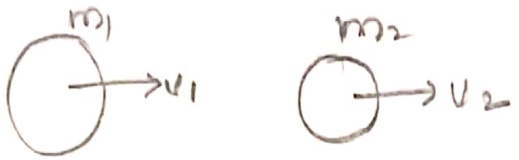
④ In case of Inelastic collisions,

$K.E_{system}$ is not conserved at all times.

One Dimensional Elastic Collision:



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[Just after collision]

① L.C.L.M,

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$m_1(u_1 - v_1) = m_2(v_2 - u_2) \quad -①$$

② L.C. of Kinetic energy,

$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$$

$$m_1(u_1^2 - v_1^2) = m_2(v_2^2 - u_2^2) \quad -②$$

$$\frac{eq\,②}{eq\,①}, \quad \frac{m_1(u_1^2 - v_1^2)}{m_1(u_1 - v_1)} = \frac{m_2(v_2^2 - u_2^2)}{m_2(v_2 - u_2)}$$

$$u_1 + v_1 = u_2 + v_2 \Rightarrow \boxed{u_1 - u_2 = v_2 - v_1}^{*} \quad -③$$

i.e, relative velocity of approach of bodies along line of impact before collision is equal to relative velocity of seperation of bodies along line of impact after collision.

from eq ③; $v_2 = u_1 + v_1 - u_2$ sub in eq ①.

$$m_1(u_1 - v_1) = m_2(u_1 + v_1 - u_2 - u_2)$$

$$m_1u_1 - m_1v_1 = m_2u_1 + m_2v_1 - 2m_2u_2$$

$$(m_1 - m_2)u_1 + 2m_2u_2 = (m_1 + m_2)v_1$$

$$\boxed{v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \left(\frac{2m_2}{m_1 + m_2}\right)u_2}$$

final velocity of first body just after collision.

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Again, From eq (2), $\boxed{v_1 = v_2 - u_1 + u_2}$ substitute in eq (1)

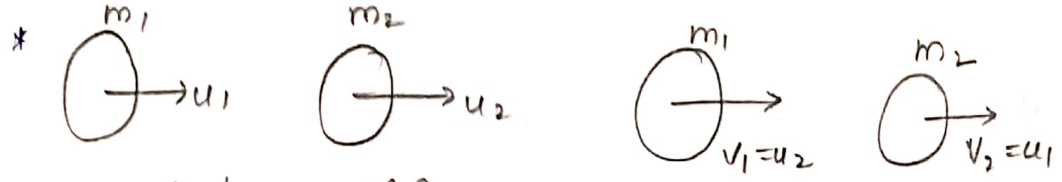
and after solving,

$$\boxed{v_2 = \left(\frac{2m_1}{m_1+m_2}\right)u_1 + \left(\frac{m_2-m_1}{m_1+m_2}\right)u_2}$$

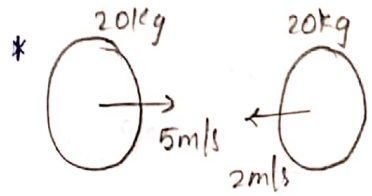
Special Cases:

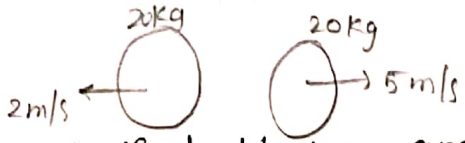
(1) If $\boxed{m_1 = m_2}$ then $\boxed{v_1 = u_2}$ and $\boxed{v_2 = u_1}$

i.e, bodies can exchange their velocity along line of impact in elastic collision for equal masses.



before collision





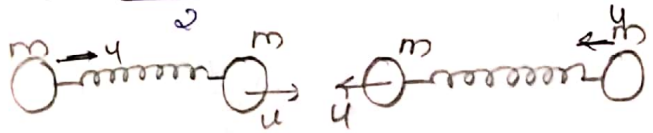
→ (2n-1) identical blocks are moving in a st. line. Odd number blocks (1, 3, 5, 7, ... 2n-1) are moving to right with 'v_0'. even number blocks (2, 4, 6, ... 2n) moving to left with 'v_0'. Find the total number of collisions?



$$N = n + (n-1) + (n-2) + \cdots + 3 + 2 + 1$$

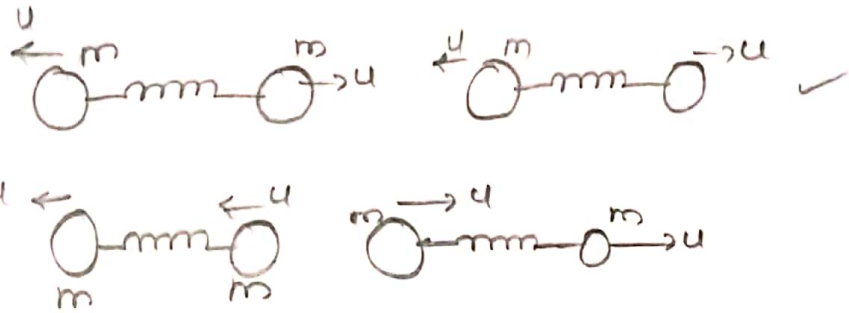
$$= \frac{n(n+1)}{2}.$$

→



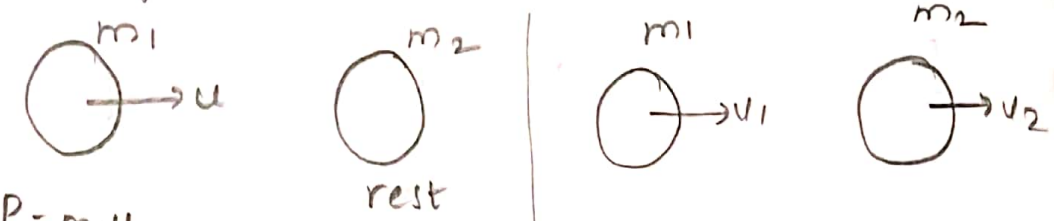


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Total no. of collisions = 2.

(2)



$P = m_1 u$

$K = \frac{1}{2} m_1 u^2$ [before collision]

* Velocity of bodies after collision,

$$V_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) u \quad , \quad V_2 = \left(\frac{2m_1}{m_1 + m_2}\right) u$$

* K.E retained by m_1 after collision,

$$K_1 = \frac{1}{2} m_1 V_1^2 = \frac{1}{2} m_1 \left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 u^2$$

$$K_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 (K)$$

* fraction of K.E retained by m_1 is,

$$\frac{K_1}{K} = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2$$

* % of K.E retained by m_1 is,

$$\frac{K_1}{K} \times 100 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 \times 100\%$$

* K.E transfered from m_1 to m_2 is,

$$K_2 = K - K_1 = \frac{1}{2} m_2 V_2^2 = \frac{1}{2} m_2 \times \frac{4 m_1^2}{(m_1 + m_2)^2} \cdot u^2$$

$$\boxed{K_2 = \frac{4 m_1 m_2}{(m_1 + m_2)^2} K}$$

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* Fraction of K.E tranfered from m_1 to m_2 is,

$$\frac{K_2}{K} = \frac{4m_1m_2}{(m_1+m_2)^2}$$

* o/o K.E transfered from m_1 to m_2 is,

$$\frac{K_2}{K} \times 100 = \frac{4m_1m_2}{(m_1+m_2)^2} \times 100\,\%$$

* Momentum of m_1 after collision is,

$$P_1 = m_1v_1 = m_1\left(\frac{m_1-m_2}{m_1+m_2}\right)u = \left(\frac{m_1-m_2}{m_1+m_2}\right)p$$

* Fraction of momentum retained by m_1 is

$$\frac{P_1}{P} = \frac{m_1-m_2}{m_1+m_2}$$

: o/o of P.E $\Rightarrow \frac{P_1}{P} \times 100 = \left(\frac{m_1-m_2}{m_1+m_2}\right) \times 100\,\%$.

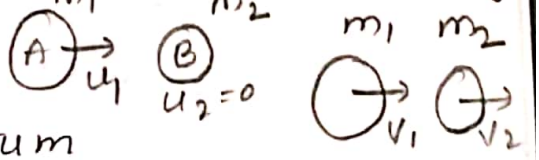
: momentum transfered from m_1 to m_2 is,

$$P_2 = P - P_1 = m_2v_2 = \frac{2m_1m_2}{m_1+m_2}u = \frac{2m_2}{m_1+m_2} \cdot p$$

* fraction of momentum transfered from m_1 to m_2 is,

$$\boxed{\frac{P_2}{P} = \frac{2m_2}{m_1+m_2}}$$

→ For a given value of mass of body A, compare mass of body B so that after collision second body recoils ① with greater speed ② with greater momentum ③ with greater K.E.



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① Velocity of second body,

let $\frac{m_2}{m_1} = K$ (say)

$$v_2 = \frac{2m_1u}{m_1+m_2}$$

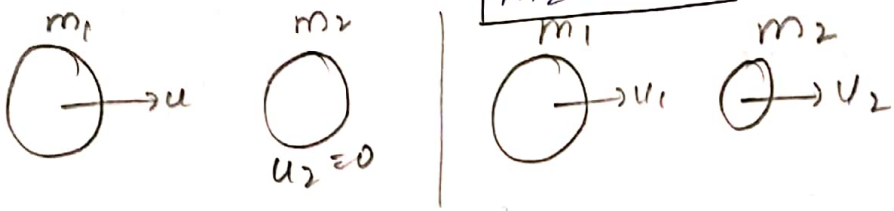
$$v_2 = \frac{2m_1u}{m_1+Km_1}$$

$$\boxed{v_2 = \frac{2u}{1+K}}$$

for $v_2 \to (v_2)_{max}$, $K \to 0$

$\frac{m_2}{m_1} \to 0$

$\boxed{m_2 << m_1}$



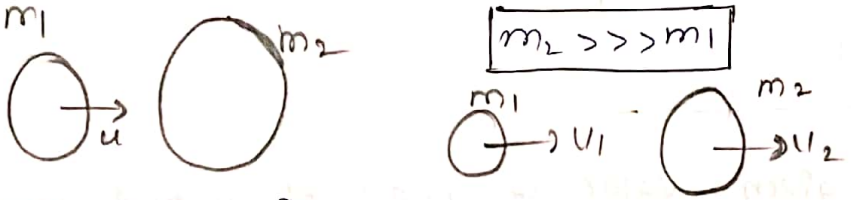
② $$P_2 = \frac{2m_1m_2}{m_1+m_2}u$$

$$P_2 = \left(\frac{2m_1Km_1}{m_1+Km_1}\right)u = \left(\frac{2K}{1+K}\right)m_1u$$

$$P_2 = \left(\frac{2}{1+\frac{1}{K}}\right)m_1u$$

$(P_2) \to (P_2)_{max}$; $1/K \to 0$

$K \to \infty$



$\boxed{m_2 >>> m_1}$

③ $$K_2 = \frac{1}{2}\frac{2m_1^2m_2}{(m_1+m_2)^2}u^2$$

$$= \frac{2m_1^2(Km_1)}{(m_1+Km_1)^2}u^2$$

$$= \frac{2Km_1u^2}{(1+K)^2} = \frac{2Km_1u^2}{(K-1)^2+4K}$$

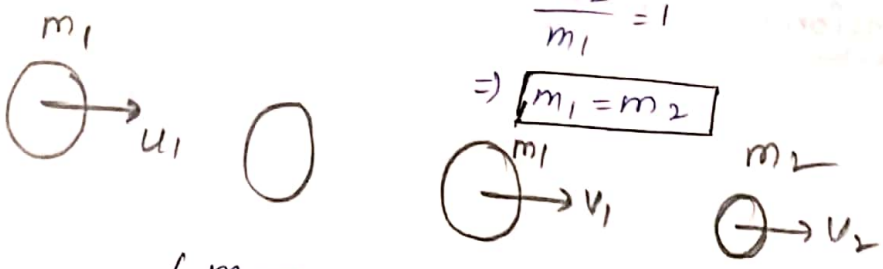
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$K_2 \longrightarrow (K_2)_{max}$

when $K = 1$

$$\frac{m_2}{m_1} = 1$$

$$\Rightarrow \boxed{m_1 = m_2}$$

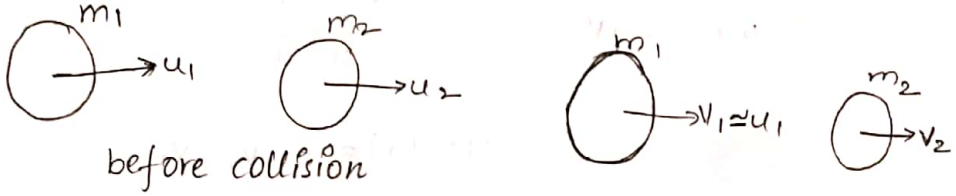


(3)

$$V_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_1 + \left(\frac{2m_2}{m_1 + m_2}\right) u_2$$

$$V_2 = \left(\frac{2m_1}{m_1 + m_2}\right) u_1 + \left(\frac{m_2 - m_1}{m_1 + m_2}\right) u_2$$

(a) If $m_1 >> m_2$ then



before collision

* $V_1 \simeq u_1$

& $V_2 = \frac{m_1}{m_2}\left(\frac{2}{1 + \frac{m_2}{m_1}}\right) u_1 + \frac{m_1\left(\frac{m_2}{m_1} - 1\right)}{m_1\left(\frac{m_2}{m_1} + 1\right)} u_2 \Rightarrow \boxed{V_2 = 2u_1 - u_2}$

If $u_2 = 0$ then,

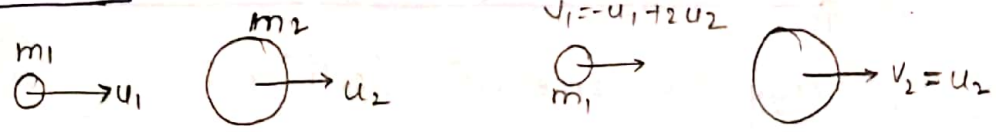
$$\boxed{V_2 \simeq 2u_1}$$

(b) If $m_1 << m_2$ then

$$V_1 = \frac{m_2\left(\frac{m_1}{m_2} - 1\right)}{m_2\left(\frac{m_1}{m_2} + 1\right)} u_1 + \frac{m_2}{m_2}\left(\frac{2}{\frac{m_1}{m_2} + 1}\right) u_2$$

$$\boxed{V_1 = -u_1 + 2u_2} \quad *$$

$$\boxed{V_2 \simeq +u_2}$$



before collision

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* $u_2 = 0$

$\Rightarrow \boxed{v_1 = -u_1}$ & $\boxed{v_2 = 0}$

One Dimensional Inelastic Collision :



before collision

after collision

* L.C.L.M,

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$m_1(u_1 - v_1) = m_2(v_2 - u_2) \quad \text{—①}$$

we have, $e = \frac{v_2 - v_1}{u_1 - u_2}$

$$(u_1 - u_2)e = v_2 - v_1$$

$$v_2 = v_1 + e(u_1 - u_2) \quad \text{—②}$$

Solve eq ② in ①

$$m_1(u_1 - v_1) = m_2(v_1 + e(u_1 - u_2) - u_2)$$

$$m_1u_1 - m_1v_1 = m_2v_1 + em_2u_1 - em_2u_2 - m_2u_2$$

$$(m_1 - em_2)u_1 + m_2(1+e)u_2 = (m_1 + m_2)v_1$$

$$\boxed{v_1 = \left(\frac{m_1 - em_2}{m_1 + m_2}\right)u_1 + \frac{m_2(1+e)}{m_1 + m_2}u_2} \quad *$$

Similarly,

$$\boxed{v_2 = \frac{m_1(1+e)}{m_1 + m_2}u_1 + \left(\frac{m_2 - em_1}{m_1 + m_2}\right)u_2}$$

Loss in Kinetic Energy in Inelastic Collision :

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$K.E_i - K.E_f = \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 - \frac{1}{2}m_1v_1^2 - \frac{1}{2}m_2u_2^2$

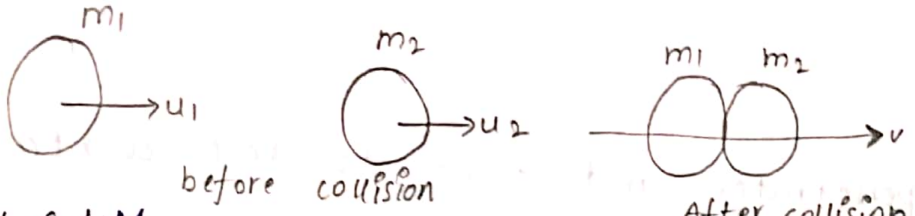
* $\boxed{\Delta K = \frac{1}{2}\frac{m_1m_2}{m_1+m_2}(\bar{u}_1-\bar{u}_2)^2(1-e^2)}$

* For any collision,

① L.C.L.M ② $e = \frac{|v_2-v_1|}{|u_1-u_2|}$

use these to get answers.

Perfectly inelastic collision:



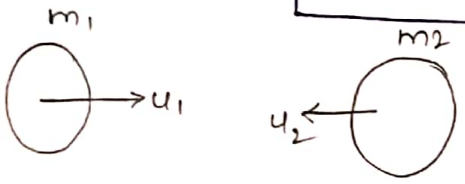
* L.C.L.M,

$m_1\bar{u}_1 + m_2\bar{u}_2 = (m_1+m_2)\bar{v}$

$m_1u_1\uparrow + m_2u_2\uparrow = (m_1+m_2)\bar{v}$

$\boxed{\bar{v} = \left(\frac{m_1u_1+m_2u_2}{m_1+m_2}\right)\uparrow}$

*



Here, e = 0

$\bar{v} = \frac{m_1u_1\uparrow - m_2u_2\uparrow}{m_1+m_2} = \left(\frac{m_1u_1-m_2u_2}{m_1+m_2}\right)\uparrow$

$\boxed{v=0}$ when $\boxed{m_1u_1 = m_2u_2}$

*



$\boxed{\tan\theta = \frac{m_2u_2}{m_1u_1}}$



$m_1u_1\uparrow + m_2u_2\uparrow = (m_1+m_2)\bar{v}$

$\bar{v} = \frac{m_1u_1\uparrow + m_2u_2\uparrow}{m_1+m_2}$

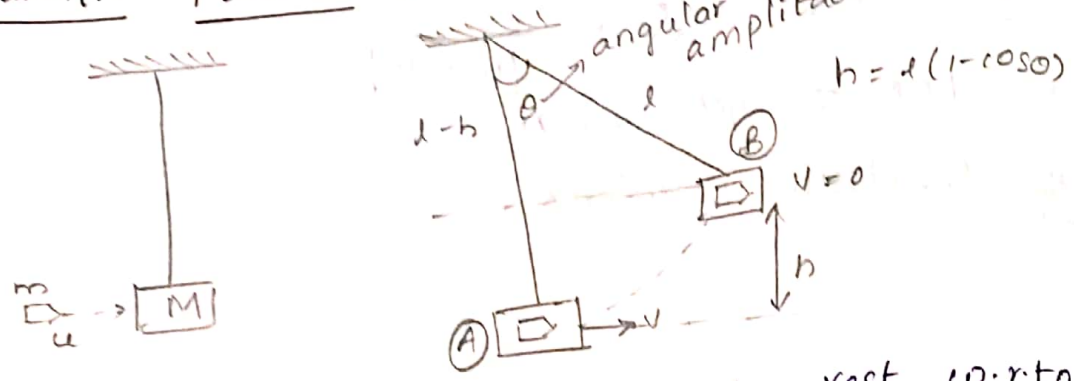
$|\bar{v}| = \frac{\sqrt{m_1^2u_1^2+m_2^2u_2^2}}{m_1+m_2}$

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* Loss in K.E is maximum in perfectly inelastic collision.

$$\Delta K = \frac{1}{2}\frac{m_1 m_2}{m_1+m_2}(\bar{u}_1-\bar{u}_2)^2$$

Ballistic Pendulum :



① Bullet penetrates and comes to rest w.r.to block;

$$mu = (m+M)V$$

$$V = \frac{mu}{m+M} \quad *$$

from K.C.E,

$$\frac{1}{2}(m+M)V^2 = (m+M)gh$$

$$h = \frac{V^2}{2g}$$

$$l(1-\cos\theta) = \frac{V^2}{2g}$$

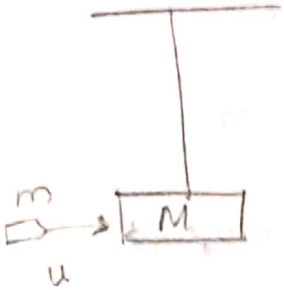
$$1-\cos\theta = \frac{V^2}{2gl}$$

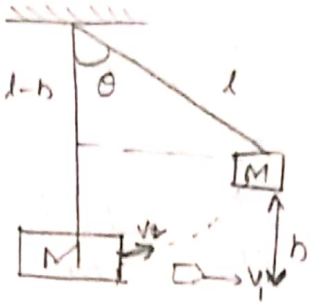
$$\cos\theta = 1-\frac{V^2}{2gl}$$

$$\theta = \cos^{-1}\left(1-V^2/2gl\right)$$

② Bullet comes out;

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$mu = MV_2 + mV_1$ —— (1)

for ballastic pendulum,

$$\frac{1}{2} M V_2^2 = Mgh$$

$$h = \frac{V_2^2}{2g}$$

$$l(1-\cos\theta) = \frac{V_2^2}{2g}$$

$$\cos\theta = 1 - \frac{V_2^2}{2gl}$$

$$\boxed{\theta = \cos^{-1}\left(1 - \frac{V_2^2}{2gl}\right)}$$

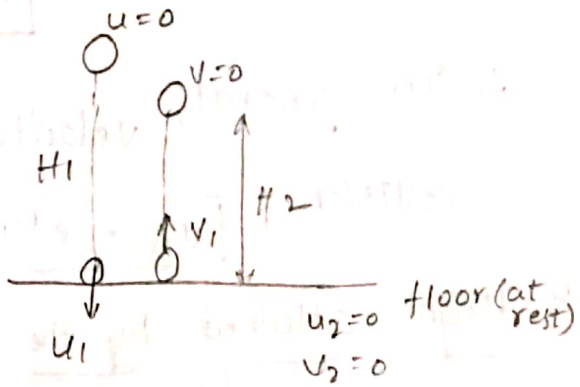
Applications :

(1) To find coefficient of restitution :

Here $\boxed{u_1 = \sqrt{2gH_1}}$

$$H_2 = \frac{V_1^2}{2g}$$

$$\boxed{V_1 = \sqrt{2gH_2}}$$



now, $e = \dfrac{\overline{V_2} - \overline{V_1}}{\overline{u_1} - \overline{u_2}}$

$$e = \frac{-\overline{V_1}}{\overline{u_1}} = \frac{-V_1 \uparrow}{u_1(-\uparrow)} = \frac{V_1}{u_1} = \frac{\sqrt{2gH_2}}{\sqrt{2gH_1}} = \sqrt{\frac{H_2}{H_1}}$$

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$$e = \frac{v_1}{u_1} = \sqrt{\frac{H_2}{H_1}} = \frac{t_2}{t_1}$$

$H_1 = \frac{1}{2} g t_1^2$

$H_1 \propto t_1^2$

$H_2 = \frac{1}{2} g t_2^2$

(2) Velocity * $F_{avg} = \frac{\Delta P}{\Delta t} = \frac{P_f - P_i}{\Delta t} = \frac{m v_1 \uparrow - m u_2 (-\uparrow)}{\Delta t}$ (on ball)

$= \frac{m(v_1 + u_1)\uparrow}{\Delta t}$ ↓ time of contact of ball with floor.

$= \frac{m(\sqrt{2gH_1} + \sqrt{2gH_2})\uparrow}{\Delta t}$

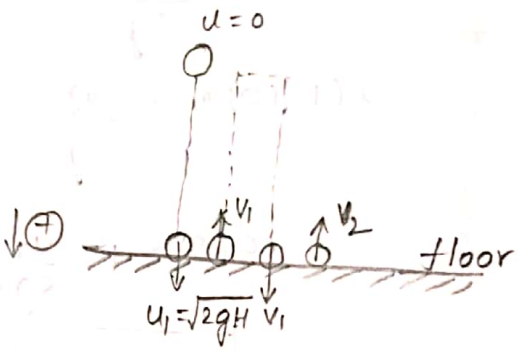
(2)** Velocity of Body after n^{th} collision:

we have,

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

$$e = \frac{-v_1}{u_1}$$

$$\boxed{v_1 = -eu_1}$$



after first collision, $v_1 = eu_1 \ (\uparrow)$

after second collision, $v_2 = ev_1 = e(eu_1)$

$$\boxed{v_2 = e^2 u_1}$$

$v_1 + v_2 + v_3 + \cdots$
$= u + eu + e^2 u + \cdots$
$= u\left(\frac{1}{1-e}\right) = \frac{u}{1-e}$

∴ In general, velocity of body after n^{th} collision, $\boxed{v_n = e^n u_1}$ (along line of impact)

(3) Height attained by the body during n^{th} collision:

we have,

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

$$e = \frac{-\sqrt{2gH_1}\uparrow}{\sqrt{2gH}\,(-\uparrow)}$$

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$\Rightarrow$ $H_1 = e^2 H$; after 1st collision,

Again,

$$H_2 = \frac{v_2^2}{2g}$$

$$= \frac{(e^2 u_1)^2}{2g}$$

$$H_2 = \frac{e^4 \cdot u_1^2}{2g} = e^4 \cdot H$$

$$H_2 = e^{2\cdot 2} H$$

similarly, $H_3 = e^{2\cdot 3} H = e^6 H$

$\therefore$ In general, height attained by body after n^{th} collision, $\boxed{H_n = e^{2n} H}$

(4) Total distance travelled by the body before coming to rest :

$$H_T = H + H_1 + H_1 + H_2 + H_2 + H_3 + H_3 + \cdots$$

$$= H + H_1 + H_1 + 2H_2 + 2H_3 + \cdots$$

$$= H + 2e^2 H + 2e^4 H + 2e^6 H + \cdots$$

$$= H + 2e^2 H (1 + e^2 + e^4 + \cdots)$$

$$= H + 2e^2 H \left(\frac{1}{1-e^2}\right)$$

$$= H \left(1 + \frac{2e^2}{1-e^2}\right)$$

$$\boxed{H_T = H\left(\frac{1+e^2}{1-e^2}\right)}$$

(5) Total time taken by the body before coming to rest :

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$$T = t_0 + t_1 + t_1 + t_2 + t_2 + t_3 + t_3 + \cdots$$

$$= \sqrt{\frac{2H}{g}} + 2\sqrt{\frac{2H_1}{g}} + 2\sqrt{\frac{2H_2}{g}} + \cdots$$

$$= \sqrt{\frac{2H}{g}} + 2\sqrt{\frac{2e^2H}{g}} + 2\sqrt{\frac{2e^4H}{g}} + \cdots$$

$$= \sqrt{\frac{2H}{g}}\left[1 + 2e + 2e^2 + 2e^3 + \cdots\right]$$

$$= \sqrt{\frac{2H}{g}}\left[1 + 2e\left(1 + e + e^2 + \cdots\right)\right]$$

$$= \sqrt{\frac{2H}{g}}\left[1 + \frac{2e}{1-e}\right]$$

$$\boxed{T = \sqrt{\frac{2H}{g}}\left[\frac{1+e}{1-e}\right]}$$

* Time taken by the body before making 2nd collision;

$$T = t + 2t_1$$

$$= \sqrt{\frac{2H}{g}} + 2\sqrt{\frac{2H_1}{g}}$$

$$= \sqrt{\frac{2H}{g}} + 2\sqrt{\frac{2e^2H}{g}}$$

$$\boxed{T = \sqrt{\frac{2H}{g}}\left(1 + 2e\right)}$$

$$T = t + 2t_1 + 2t_2$$

$$= \sqrt{\frac{2H}{g}} + 2\sqrt{\frac{2e^2H}{g}} + 2\sqrt{\frac{2e^4H}{g}}$$

$$= \sqrt{\frac{2H}{g}}\left(1 + 2e + 2e^2\right)$$

⑥ Average speed:

$$V_{avg} = \frac{\text{Total distance travelled}}{\text{total time taken}}$$

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$$V_{avg} = \frac{H\left[\frac{1+e^2}{1-e^2}\right]}{\sqrt{\frac{2H}{g}}\left[\frac{1+e}{1-e}\right]}$$

$$\boxed{V_{avg} = H\sqrt{\frac{g}{2H}}\left[\frac{1+e^2}{(1+e)^2}\right]} \qquad V_{avg} = \sqrt{\frac{Hg}{2}}\left(\frac{1+e^2}{(1+e)^2}\right)$$

(7) Average Velocity:

$$V'_{avg} = \frac{\text{net displacement}}{\text{total time}}$$

$$= \frac{H}{\sqrt{\frac{2H}{g}}\left(\frac{1+e}{1-e}\right)}$$

$$\boxed{V'_{avg} = \sqrt{\frac{gH}{2}}\left(\frac{1-e}{1+e}\right)}$$

(8) change in Momentum:

* change in momentum during 1st collision,

$$\Delta\bar{P}_1 = m(v_1+u)\uparrow = m(eu+u) = mu(1+e)\uparrow$$

* during 2nd collision,

$$\Delta\bar{P}_2 = m(v_2+v_1)\uparrow = m(e^2u+eu) = meu(1+e)\uparrow$$

∴ Total change in momentum,

$$\Delta\bar{P} = \Delta\bar{P}_1 + \Delta\bar{P}_2 + \text{- - - -}$$

$$= mu(1+e) + meu(1+e) + me^2u(1+e) + \text{- - -}$$

$$= mu(1+e)\left[1+e+e^2+\text{- - -}\right]$$

$$\# \quad \frac{mu(1+e)}{1-e} = mu\left(\frac{1+e}{1-e}\right) = mu(1+e)\left[1+2e(1+e+\text{- -})\right]$$

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$$= mu(1+e)\left[1+\frac{2e}{1-e}\right]$$

$$= \frac{mu(1+e)^{\cancel{2}}}{1-e}$$

$$\boxed{\Delta P = m\sqrt{2gH}\left[\frac{1+e}{1-e}\right](\uparrow)}$$

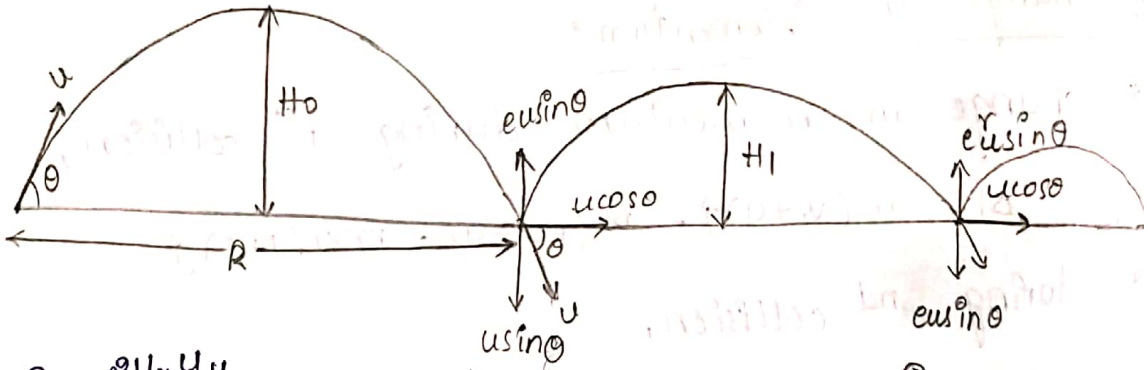
∴ Avg force on ball,

$$F_{avg} = \frac{\Delta P}{\Delta t} = \frac{m\sqrt{2gH}\left[\frac{1+e}{1-e}\right]}{\sqrt{\frac{2H}{g}}\left[\frac{1+e}{1-e}\right]}$$

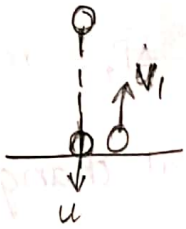
$$\boxed{F_{avg} = mg}$$

In case of projectile Motion:

(all surfaces are smooth and neglecting air resistance)



$$R = \frac{2u_x u_y}{g}\ ;\ H = \frac{u_y^{\cancel{2}}}{2g}\ ;\ T = \frac{2u_y}{g}$$



$F = 0$

$\Rightarrow$ $P = \text{const}$

$mu_1 = mv_1$

$$\boxed{u_1 = v_1}$$

$R = R_0 + R_1$

$= R_0 + eR_0$

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Total horizontal distance:

$R = R_0 + R_1 + R_2 + R_3 + \dots$

$= R_0 + R_0 e + e^2 R_0 + \dots$

$= R_0 (1 + e + e^2 + \dots)$

$= R_0 \left(\frac{1}{1-e}\right)$

Total maximum height

$H = H_0 + H_1 + H_2 + \dots$

$= \frac{u_y^2}{2g} + \frac{(eu_y)^2}{2g} + \frac{(e^2 u_y)^2}{2g} + \dots$

$= \frac{u_y^2}{2g} \left[1 + e^2 + e^4 + \dots\right]$

$= \frac{u^2}{2g} \left[\frac{1}{1-e^2}\right] = \frac{H}{1-e^2}$

Total time taken before coming to rest:

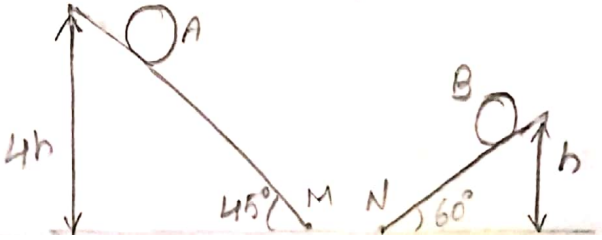
$T = T_0 + T_1 + T_2 + \dots$

$= \frac{2u\sin\theta}{g} + \frac{2eu\sin\theta}{g} + \frac{2e^2 u\sin\theta}{g} + \dots$

$= \frac{2u\sin\theta}{g} (1 + e + e^2 + \dots)$

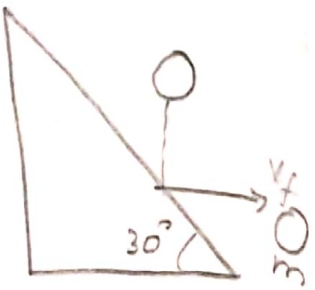
$= \frac{2u\sin\theta}{g(1-e)} = \frac{T}{1-e}$

→ 1) Two identical balls A and B are released from the positions shown in figure. They collide elastically on horizontal portion MN. All surfaces are smooth. The ratio of heights attained by A & B after collision will be a) 1:4 b) 2:1 c) 4:13 d) 2:5



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2) As shown in fig, A body of mass m moving vertically down with speed 3m/s hits a smooth fixed plane and rebounds with velocity V_f in the horizontal direction. If angle of inclination is 30°, the velocity V_f will be a) 3m/s b) $\sqrt{3}$ m/s c) $1/\sqrt{3}$ m/s d) Not possible.



3) A sphere strikes a ball and rebounds with coefficient of restitution $\frac{1}{3}$. If it rebounds with velocity of 0.1 m/s at an angle 60° to the normal to the wall. The loss in K.E is a) 50% b) 33 1/3% c) 40% d) 66 2/3%

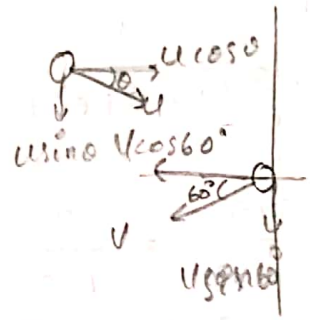
2)

$$v_f = e u \cot\theta$$

$$= u \tan\theta$$

$$= 3 \times \frac{1}{\sqrt{3}} = \sqrt{3} \text{ m/s}$$

3)



$$u \sin\theta = V \sin 60°$$

$$V\cos 60° = -e u \cos\theta$$

$$(3V\cos 60°)^2 + (V \sin 60°)^2 = u^2$$

$$\left(3(0.1)\left(\frac{1}{2}\right)\right)^2 + \left(0.1 \times \frac{\sqrt{3}}{2}\right)^2 = u^2$$

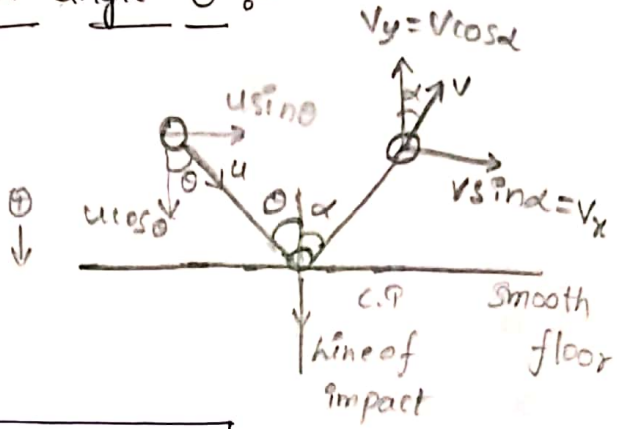
$$\frac{9}{4} \times \frac{1}{100} + \frac{1}{100} \times \frac{3}{4} = v^2$$

$$\frac{3}{100} = u^2$$

$$\text{Loss in K.E} = \frac{1}{2} m u^2 - \frac{1}{2} m v^2 = \frac{1}{2} \times m \left(\frac{3}{100} - \frac{1}{100}\right)$$

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Body strikes the floor at an angle θ :-



* Impulse eqn :

$$\bar{P}_i \pm \int F \cdot dt = \bar{P}_f$$

$$mu\cos\theta - \int N \cdot dt = -mv\cos\alpha$$

$$\boxed{\int N \cdot dt = mu\cos\theta + mv\cos\alpha}$$

* Along common tangent,

$$mu\sin\theta - \int F_H \cdot dt = mv\sin\alpha$$

$$mu\sin\theta = mv\sin\alpha$$

$$\boxed{u\sin\theta = v\sin\alpha} \quad -①$$

* coefficient of restitution :

$$e = \frac{\bar{V}_2 - \bar{V}_1}{\bar{u}_1 - \bar{u}_2} \quad \text{(along L.O.I)}$$

$$= \frac{-\bar{V}_1}{\bar{u}_1}$$

$$= \frac{+(+v\cos\alpha)}{+u\cos\theta}$$

$$e = \frac{v\cos\alpha}{u\cos\theta}$$

$$\boxed{v\cos\alpha = eu\cos\theta} \quad -②$$

$$\frac{\text{eq ①}}{\text{eq ②}}, \quad \frac{u\sin\alpha}{v\cos\alpha} = \frac{u\sin\theta}{eu\cos\theta}$$

$$\boxed{\tan\alpha = \frac{\tan\theta}{e}}$$

and $\quad v^2\sin^2\alpha + v^2\cos^2\alpha = u^2\sin^2\theta + e^2u^2\cos^2\theta$

$$v^2 = u^2(\sin^2\theta + e^2\cos^2\theta)$$

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$$\boxed{v = u\sqrt{\sin^2\theta + e^2\cos^2\theta}}$$

Points:

① loss in K.E of ball,

$$\Delta K = K.E_i - K.E_f$$

$$= \frac{1}{2}mu^2 - \frac{1}{2}mv^2$$

% loss in K.E $\Rightarrow \frac{\Delta K}{K_i} \times 100 = \frac{\frac{1}{2}mu^2 - \frac{1}{2}mv^2}{\frac{1}{2}mu^2} \times 100$

$$= \frac{u^2 - v^2}{u^2} \times 100$$

② For elastic collision, $\boxed{e = 1}$

$\Rightarrow \tan\alpha = \tan\theta \Rightarrow \boxed{\theta = \alpha}$

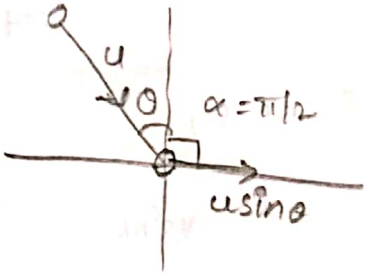
& $\boxed{v = u}$, $\boxed{\Delta K = 0}$

③ For perfectly inelastic collision,

$\boxed{e = 0}$

$\Rightarrow \tan\alpha = \frac{\tan\theta}{0} =$ infinity

$\boxed{\alpha = \pi/2}$



& $\boxed{v = u\sin\theta}$

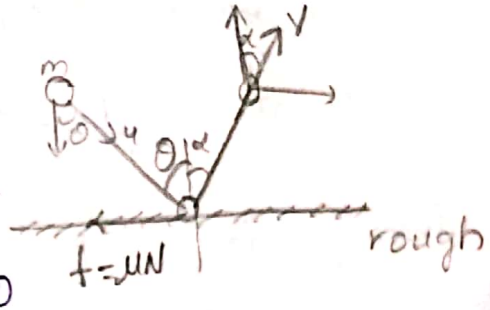
$$\frac{\Delta K}{K} = \frac{\frac{1}{2}mu^2 - \frac{1}{2}mu^2\sin^2\theta}{\frac{1}{2}mu^2} \times 100$$

$$= \cos^2\theta \times 100\ \%$$

Impulse eqn:

$$mu\cos\theta - \int N\,dt = -mv\cos\alpha$$

$$\boxed{\int N\,dt = mu\cos\theta + mv\cos\alpha} \quad \text{—①}$$



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* Along common tangent,

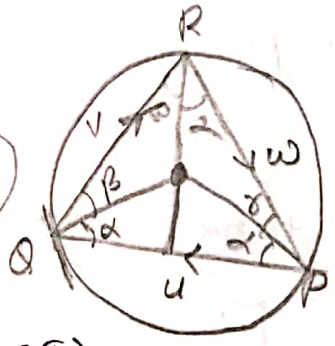
$$mu\sin\theta - \int \mu N dt = mv\sin\alpha$$

$$mu\sin\theta - \mu[ma\cos\theta + mv\cos\alpha] = mv\sin\alpha \quad -(2)$$

$$e = \frac{\overline{v_2} - \overline{v_1}}{\overline{u_1} - \overline{u_2}} \quad -(3)$$

→ A smooth ring is kept on a smooth horizontal surface from a point P of the ring, a particle is projected at an angle 'α' to the radius vector at P. if 'e' is the coeff of restitution bfw ring & particle. Show that the particle will return to the point of projection after two reflections if $\cot\alpha = \frac{1}{e} + \frac{1}{e^2} + \frac{1}{e^3}$.

$$\cot^2\alpha = \frac{1}{e} + \frac{1}{e^2} + \frac{1}{e^3}$$



(1) at Q,

$$v\sin\beta = u\sin\alpha \quad -(1) \text{ (along C.T)}$$

$$v\cos\beta = eu\cos\alpha \quad -(2) \text{ (along CN)}$$

$$\Rightarrow \tan\beta = \frac{\tan\alpha}{e} \quad -(3)$$

at R,

$$v\sin\beta = \omega\sin\gamma \text{ (along C.T)}$$

$$\omega\cos\gamma = ev\cos\beta \text{ (along C.N)}$$

$$\tan\gamma = \frac{\tan\beta}{e} = \frac{\tan\alpha}{e^2} \quad -(4)$$

from figure, $2\alpha + 2\beta + 2\gamma = 180$

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$\alpha+\beta+\gamma = 90°$

$\tan(\alpha+\beta+\gamma) = \tan 90°$

$\Rightarrow \dfrac{S_1 - S_3}{1 - S_2} = \dfrac{1}{0} \Rightarrow S_2 = 1$

$\tan\alpha\tan\beta + \tan\beta\tan\gamma + \tan\gamma\tan\alpha = 1$

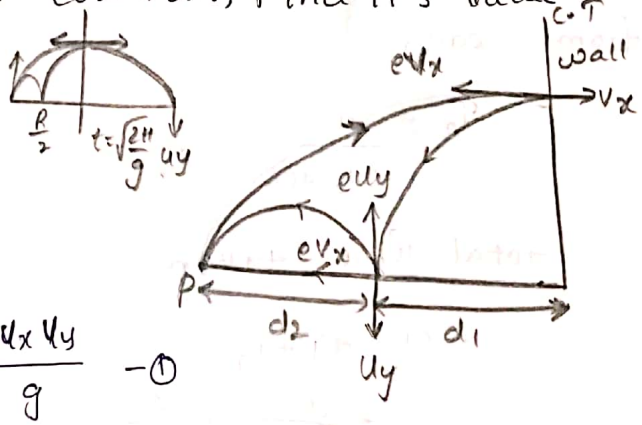
$\dfrac{\tan\alpha}{e}\cdot\dfrac{\tan\alpha}{e} + \dfrac{\tan\alpha}{e}\cdot\dfrac{\tan\alpha}{e^2} + \dfrac{\tan\alpha}{e^2}\cdot\dfrac{\tan\alpha}{} = 1$

$\tan^2\alpha\left(\dfrac{1}{e} + \dfrac{1}{e^3} + \dfrac{1}{e^2}\right) = 1$

$$\boxed{\cot^2\alpha = \frac{1}{e} + \frac{1}{e^2} + \frac{1}{e^3}}$$

$\therefore$ Hence proved.

→ A small ball is projected from point P on floor towards the wall as shown. It hits the wall when its velocity is horizontal, ball reaches point P after one bounce on the floor. If coeff of restitution is same for two collisions, find it's value.



$d_1 = (ev_x)\left(\frac{T}{2}\right)$

$= (ev_x)\left(\dfrac{u_y}{g}\right)$

$= \dfrac{ev_x u_y}{g} = \dfrac{eu_x u_y}{g}$ — ①

$d_2 = \dfrac{2(eu_x)(eu_y)}{g} = \dfrac{2e^2 u_x u_y}{g}$ — ②

from figure, $\dfrac{R}{2} = d_1 + d_2$

$\dfrac{2u_x u_y}{2g} = \dfrac{eu_x u_y}{g} + \dfrac{2e^2 u_x u_y}{g}$

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$$2e^2 + e - 1 = 0$$

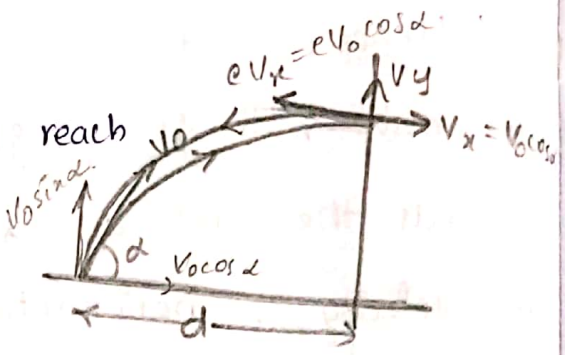
$$2e^2 + 2e - e - 1 = 0$$

$$2e(e+1) - (e+1) = 0$$

$$(e+1)(2e-1) = 0$$

$$\boxed{e = 1/2}$$

→ An inelastic ball is projected with velocity $V_0 = \sqrt{gd}$ at angle α to the horizontal towards a wall of distance d from the point of projection after collision the ball returns to the point of projection. Find e.



Time taken by body to reach the wall,

$$T_1 = \frac{d}{V_0\cos\alpha} \quad -(1)$$

Time taken by ball to reach the point of projection from wall,

$$T_2 = \frac{d}{eV_0\cos\alpha} \quad -(2)$$

∴ total time taken,

$$T = T_1 + T_2$$

$$= \frac{d}{V_0\cos\alpha} + \frac{d}{eV_0\cos\alpha}$$

$$= \frac{d}{V_0\cos\alpha}\left(\frac{e+1}{e}\right)$$

$$\boxed{T = \frac{d(1+e)}{eV_0\cos\alpha}} \quad -(3)$$

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Here vertical component of velocity is not changing while hitting the wall.

$$\text{Total time} = \frac{2V_0 \sin\alpha}{g} \quad -(4)$$

from (3) & (4)

$$\frac{d(1+e)}{eV_0\cos\alpha} = \frac{2V_0\sin\alpha}{g}$$

$$d\left(\frac{1}{e}+1\right) = \frac{2V_0^2\cos\alpha\sin\alpha}{g}$$

$$\frac{1}{e}+1 = \frac{V_0^2\sin 2\alpha}{gd}$$

$$\frac{1}{e} = \frac{V_0^2\sin 2\alpha}{gd} - 1$$

$$\boxed{e = \frac{gd}{V_0^2\sin 2\alpha - gd}}$$

$$e = \frac{gd}{gh\sin 2\alpha - gd}$$

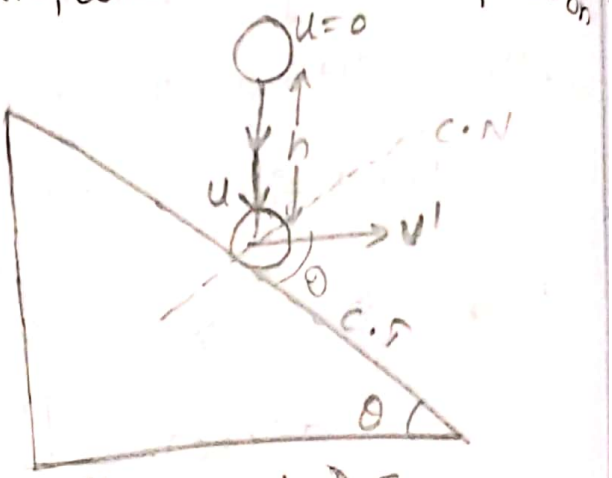
$$\boxed{e = \frac{d}{h\sin 2\alpha - d}}$$

→ A small ball is dropped from a height h on a smooth inclined plane as shown. after collision, the velocity of ball is directed horizontally. Find

1) coefficient of restitution (e)

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8) If collision is elastic, what is the impulse on the ball?

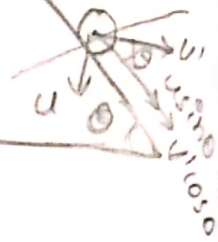


① Along C.T, velocity of ball is same, i.e,

$u\sin\theta = v'\cos\theta$

$$\boxed{v' = u\tan\theta} \quad -①$$

② $$e = \frac{\overline{V_2} - \overline{V_1}}{\overline{u_1} - \overline{u_2}} = \frac{-\overline{V_1}}{\overline{u_1}} = \frac{v'\sin\theta}{u\cos\theta}$$

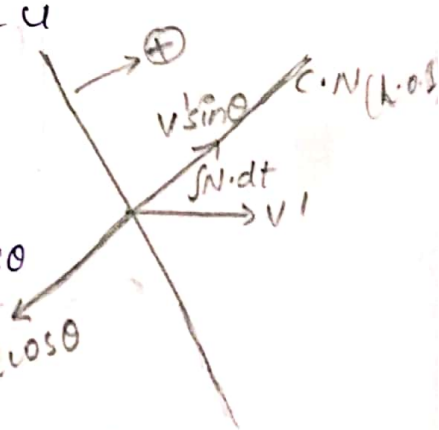


$$e = \frac{v'\tan\theta}{u} = \frac{u\tan^2\theta}{u}$$

$$\boxed{e = \tan^2\theta} \quad \text{—— [1st Ans]}$$

③ If collision is elastic,

$e = 1 \Rightarrow \theta = 45^\circ$ & $v' = u$



$$-mu\cos\theta + \int N\,dt = mv'\sin\theta$$

$$\int N\,dt = mv'\sin\theta + mu\cos\theta$$

$$= mu(\sin\theta + \cos\theta)$$

$$= mu\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right)$$

$$= mu\sqrt{2}$$

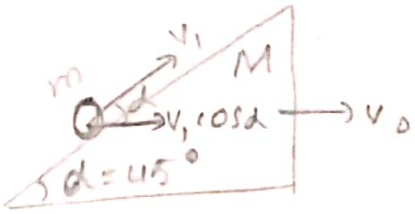
$$J = \sqrt{2}mu$$

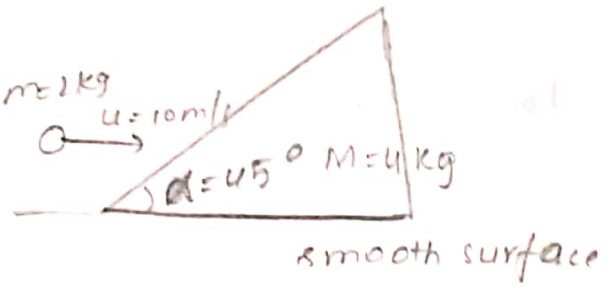
$$= \sqrt{2}m\sqrt{2gh}$$

$$\boxed{J = 2m\sqrt{gh}} \quad \text{— [2nd ans]}$$

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→ After collision the particle starts moving up the inclined plane. Calculate the velocity of wedge immediately after collision.

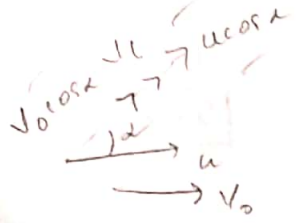




L.C.L.M,

$$mu = MV_0 + m(v_1\cos\alpha + v_0) \quad -(1)$$

along C.T,



$$u\cos\alpha = v_1 + v_0\cos\alpha$$

$$u\cos\alpha - v_0\cos\alpha = v_1$$

(1) $\Rightarrow$ $mu = MV_0 + m((u-v_0)\cos^2\alpha + v_0)$

$$2\times 10 = 4\times V_0 + 2\left((10-V_0)\left(\tfrac{1}{2}\right) + V_0\right)$$

$$10 = 2V_0 + 5 - \frac{V_0}{2} + V_0$$

$$10 - 5 = 3V_0 - \frac{V_0}{2}$$

$$5 = \frac{5V_0}{2} \Rightarrow \boxed{V_0 = 2\,m/s} \Rightarrow v_1 = (10-2)\cos\alpha$$

$$= \frac{8}{\sqrt{2}} = 4\sqrt{2}$$

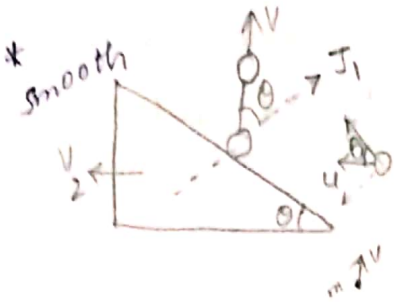
$$\overline{V_b} = \overline{V_{b/w}} + \overline{V_w}$$

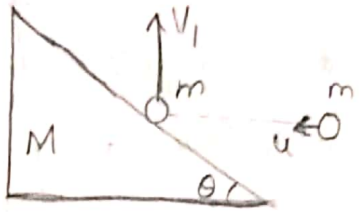
$$= v_1\cos\alpha + v_0$$

$$= 4\sqrt{2}\times\frac{1}{\sqrt{2}} + 2$$

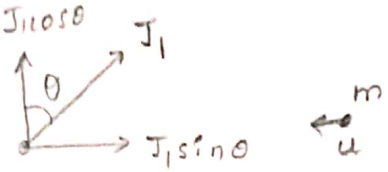
$$= 6\,m/s$$

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All surfaces are smooth

For ball, 

$mu - J_1 \sin\theta = 0$ & $0 + J_1 \cos\theta = mv_1$

$J_1 \sin\theta = mu$

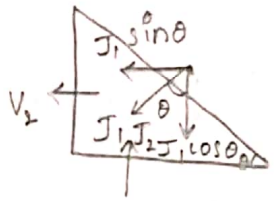
$\boxed{J_1 = mu \operatorname{cosec}\theta}$

$\boxed{J_1 = mv_1 \sec\theta}$

$\Rightarrow mu \operatorname{cosec}\theta = mv_1 \sec\theta$

$\boxed{u \cot\theta = v_1}$

On wedge,



$J_1 \sin\theta = MV_2$

$mu \operatorname{cosec}\theta \sin\theta = MV_2$

$mu = MV_2$

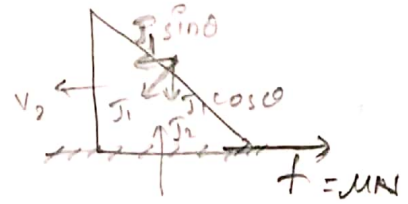
$\boxed{V_2 = \frac{mu}{M}}$

$J_2 = J_1 \cos\theta$

$= mV_2 \operatorname{cosec}\theta \cos\theta$

$= mV_2 \cot\theta$

$\boxed{J_2 = mu \cot\theta}$

* rough 

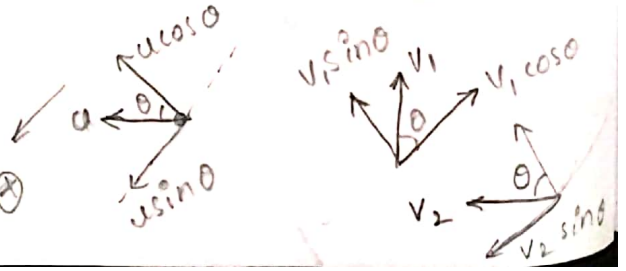
friction impulse,

$J_H = \mu J_2$

$\boxed{J_H = \mu mu \cot\theta}$

coefficient of restitution,

$e = \frac{v_2 - v_1}{u_1 - u_2}$



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$$e = \frac{v_2 \sin\theta - (-v_1 \cos\theta)}{u \sin\theta}$$

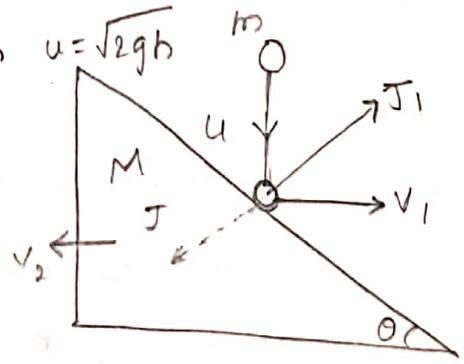
$$= \frac{v_2 + v_1 \cot\theta}{u} = \frac{\frac{mu}{M} + u \cot^2\theta}{u} = \frac{m}{M} + \cot^2\theta$$

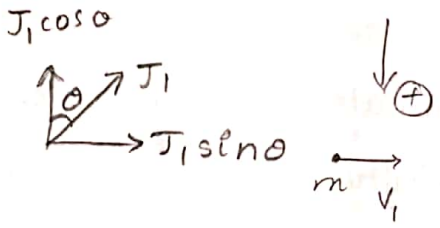
$$\boxed{e = \frac{m}{M} + \cot^2\theta}$$

Loss in K.E,

$$\Delta K = K_i - K_f = \frac{1}{2}mu^2 - \left(\frac{1}{2}mv_1^2 + \frac{1}{2}Mv_2^2\right)$$

* smooth





For ball, $mu - J_1 \cos\theta = 0$

$$J_1 \cos\theta = mu$$

$$\boxed{J_1 = mu \sec\theta}$$

& $0 + J_1 \sin\theta = mv_1$

$$J_1 = mv_1 \operatorname{cosec}\theta$$

$$mu \sec\theta = mv_1 \operatorname{cosec}\theta$$

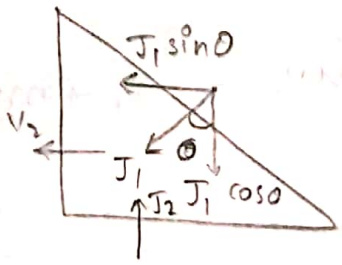
$$\boxed{v_1 = u \tan\theta}$$

For wedge,

$$J_1 \sin\theta = Mv_2$$

$$J_1 = Mv_2 \operatorname{cosec}\theta$$

$$\boxed{mv_1 = Mv_2}$$



Impulse due to ground on wedge,

$$J_2 = J_1 \cos\theta = Mv_2 \cot\theta = mu.$$

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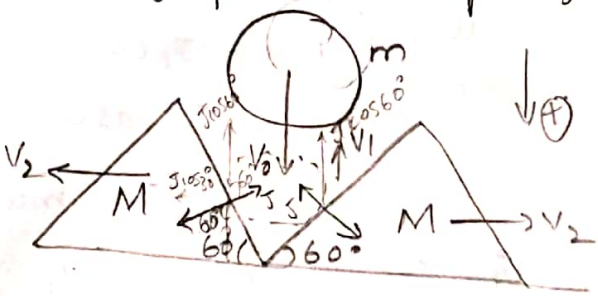
coefficient of restitution, $e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{v_2 \sin\theta - (-v_1 \sin\theta)}{u\cos\theta} = \left(\frac{v_2 + v_1}{u}\right)\tan\theta$

$= \left(\frac{mv_1 + v_1}{u}\right)\tan\theta = \frac{(m+1)\,u\tan^2\theta}{u} = \left(\frac{m}{M}+1\right)\tan^2\theta$

Loss in K.E,

$\Delta K = K_i - K_f = \frac{1}{2}mu^2 - \left(\frac{1}{2}mv_1^2 + \frac{1}{2}Mv_2^2\right)$

→ A smooth spherical ball of mass striking two identical equilateral triangular of mass M at the instant of impact. velocity of ball is v_0 taking coefficient of restitution e. determine the velocities of sphere & wedge just after collision.



On ball,

$mv_0 - 2J\cos 60 = -mv_1$

$J = m(v_0 + v_1)$ — ①

For wedge,

$J\cos 30° = Mv_2$ — ②

from ① & ②

$\frac{2}{\sqrt{3}}Mv_2 = mv_1 + mv_0$ — ③

$e = \frac{v_1\cos 60° + v_2\cos 30°}{v_0\cos 60°}$

$= \frac{v_1 + \sqrt{3}v_2}{v_0}$

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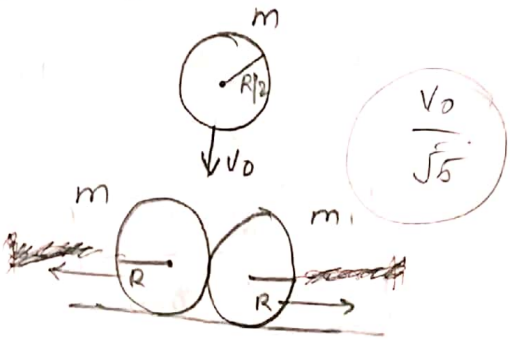
$$eV_0 = V_1 + \sqrt{3}V_2 \quad -(4)$$

from (3) & (4)

$$V_1 = \frac{(2eM - 3m)V_0}{2M+3m}$$

$$V_2 = \frac{\sqrt{3}(e+1)mV_0}{2M+3m}$$

→

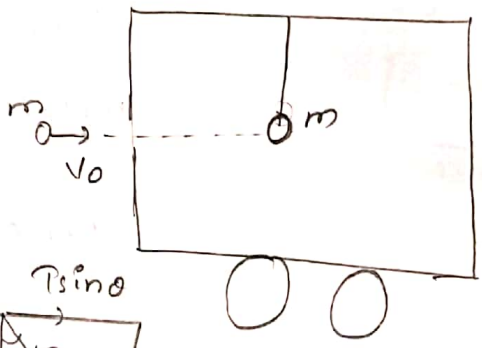


→ perfectly elastic collision

Method-I :

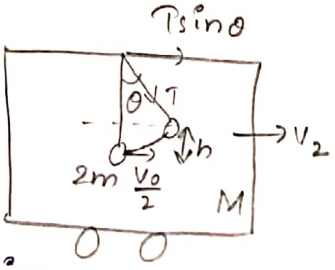
$$mV_0 = 2mv$$

$$V = \frac{V_0}{2}$$



At max deflection,

bob & trolley move with common acceleration,



L.C.L.M,

$$2m\left(\frac{V_0}{2}\right) = (2m+M)V_2$$

$$\frac{mV_0}{2m+M} = V_2$$

L.C.E,

$$\frac{1}{2}(2m)\left(\frac{V_0}{2}\right)^2 = (2m)gh + \frac{1}{2}(2m)V_2^2 + \frac{1}{2}MV_2^2$$

from this we get 'h'.

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$$\cos\theta = \frac{l-h}{l} \Rightarrow h = l(1-\cos\theta)$$

Π-method:

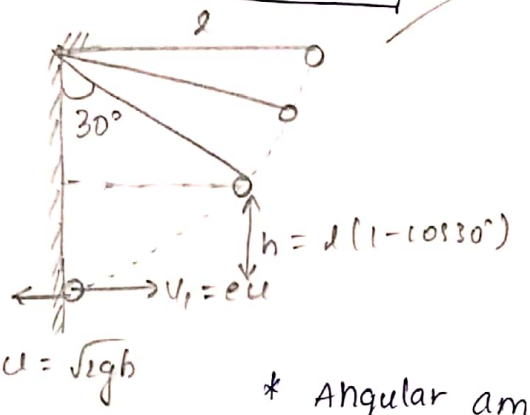
$$W_g = \Delta K_{sys/cm}$$

$$2mgh = \frac{1}{2}\frac{(2m)M}{2m+M}\left[0-\left(\frac{V_0}{2}\right)^2\right]$$

$$h = \frac{M}{8g(2m+M)}\left(\frac{V_0^2}{4}\right)$$

$$\boxed{h = \frac{MV_0^2}{8g(2m+M)}}$$

→



* Velocity after nth collision,

$$v_n = e^n u = e^n\sqrt{2gh}$$

$$K.E_n = \frac{1}{2}mv_n^2 = \frac{1}{2}me^{2n}(2gl)$$

$$= mgle^{2n}$$

* Angular amplitude is less than 30°,

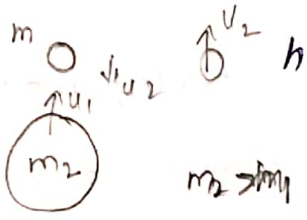
$$K.E_n < mgl(1-\cos 30°)$$

$$mgle^{2n} < mgl(1-\cos 30°)$$

$$2n\ln e < \ln\left(1-\frac{\sqrt{3}}{2}\right)$$

→





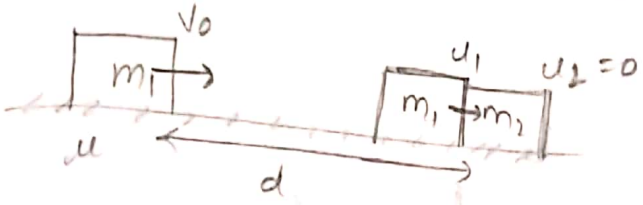
$$h = \frac{v_2^2}{2g}$$

$m_2 > m_1$

$$v_2 = \left(\frac{2m_1}{m_1+m_2}\right)u_1 + \left(\frac{m_2-m_1}{m_1+m_2}\right)u_2$$

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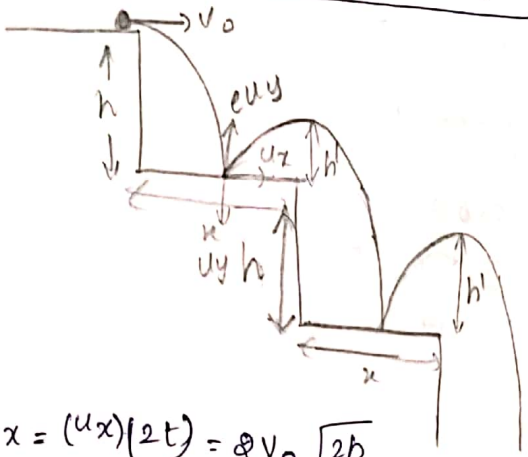
→



$$u_1^2 - v_0^2 = -2\mu g d$$

$$\boxed{u_1 = \sqrt{v_0^2 - 2\mu g d}}$$

→



$$t = \sqrt{\frac{2h}{g}}$$

$$u_y^2 - 0^2 = 2gh$$

$$u_y = \sqrt{2gh}$$

From figure,

$$-h = eu_y(2t) - \frac{1}{2}g(2t)^2$$

$$-h = e\sqrt{2gh}\left(2\sqrt{\frac{2h}{g}}\right) - \frac{1}{2}(g)(4)\left(\frac{2h}{g}\right)$$

$$-h = 2he \times 2 - 4h$$

$$-h = 4he - 4h$$

$$-1 + 4 = 4e$$

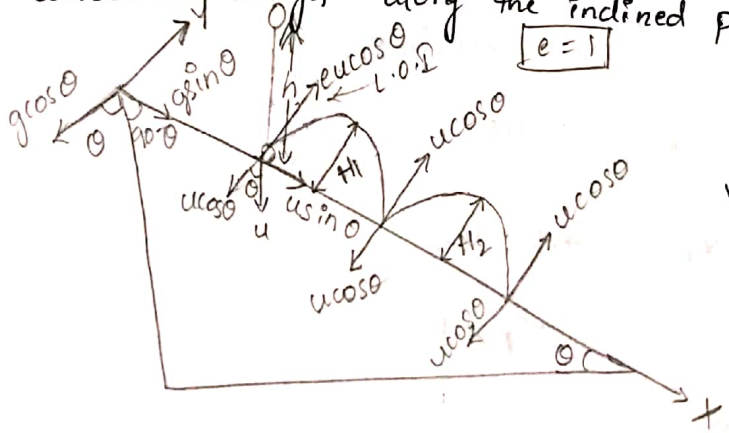
$$\boxed{\frac{3}{4} = e}$$

$$x = (u_x)(2t) = 2v_0\sqrt{\frac{2h}{g}}$$

$$\boxed{x = 2v_0\sqrt{\frac{2h}{g}}}$$

→ A small ball is dropped on inclined plane from height 'h' as shown. Assume all collisions are elastic. Find the ratio of consecutive ranges along the inclined plane.

$\boxed{e = 1}$



$$v_y^2 - u_y^2 = 2as^0$$

$$\boxed{v_y = u_y}$$

* Along X-axis, $y = 0$

we have, $y = u_y t + \frac{1}{2}a_y t^2$

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$$0 = u\cos\theta T + \frac{1}{2}(-g\cos\theta)T^2$$

$$+u = +\frac{1}{2}gT$$

$$u = \frac{1}{2}gT \Rightarrow \boxed{T = \frac{2u}{g}} = \text{const.}$$

i.e, time b/w successive collision is same.

* $0^2 - (u\cos\theta)^2 = 2(-g\cos\theta)H$.

$$u^2\cos^2\theta = 2g\cos\theta H$$

$$\boxed{H = \frac{u^2\cos\theta}{2g}} = \text{const.}$$

i.e, $H_1 = H_2 = \cdots\cdots$

*Range,

$$R = u_x t + \frac{1}{2}a_x t^2$$

$$= \left(\frac{u_x + v_x}{2}\right)T \quad (a_x = \text{const})$$

and change in velocity,

$$\Delta v_x = a_x T$$

$$\Delta v_x = (g\sin\theta)\frac{2u}{g}$$

$$\Delta v_x = 2u\sin\theta$$

i.e, $v_x - u_x = u\sin\theta$

$$v_x = u\sin\theta + u_x$$

$$\boxed{v_x = 2u\sin\theta}$$

* $R = \left(\frac{u\sin\theta + 3u\sin\theta}{2}\right)T = 2u\sin\theta T.$

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$$R_2 = \left(\frac{3u\sin\theta + 5u\sin\theta}{2}\right)T$$

$$= 4u\sin\theta\, T$$

$$R_3 = \left(\frac{5u\sin\theta + 7u\sin\theta}{2}\right)T$$

$$= 6u\sin\theta\, T.$$

$\therefore R_1 : R_2 : R_3 : \ldots = 2 : 4 : 6 : \ldots = 1 : 2 : 3 \ldots$

* If the collision is inelastic:

Max height:

$$H_1 = \frac{(eu\cos\theta)^2}{2g\cos\theta} = \frac{e u^2 \cos\theta}{2g} \quad -①$$

$$H_2 = \frac{(e^2 u\cos\theta)^2}{2g\cos\theta} = \frac{e^4 u^2 \cos\theta}{2g} = e^2 H_1$$

$$H_3 = \frac{e^6 u^2 \cos\theta}{2g} = e^4 H_1.$$

$\therefore H_1 : H_2 : H_3 : \ldots = 1 : e^2 : e^4 : \ldots$

Time of flight b/w two consecutive collisions:

$$T_1 = \frac{2eu\sin\theta}{g\cos\theta} = \frac{2eu}{g}$$

$$T_2 = \frac{2e^2 u}{g} = e T_1$$

$$T_3 = e^2 . T_1$$

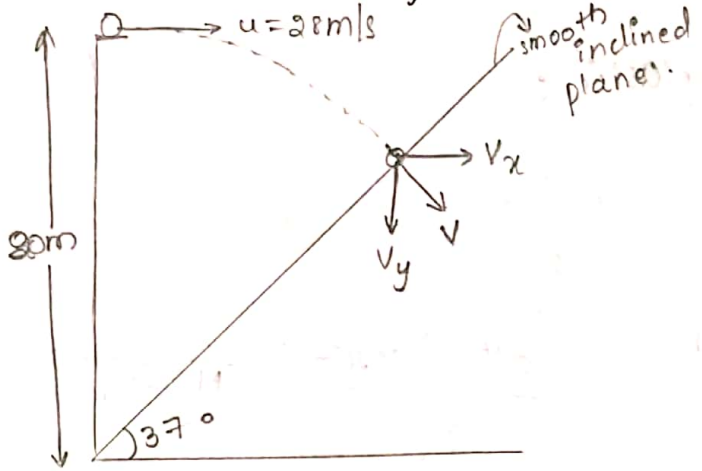
$T_1 : T_2 : T_3 : \ldots = 1 : e : e^2 : \ldots$

Range:

$$R_1 = \left(\frac{u_x + v_x}{2}\right)T$$

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Show that whatever be the coefficient of restitution b/w the ball and the incline, the ball will always hit the inclined for the 2nd time after point above B.



From figure,

$$\tan 37^\circ = \frac{H-y}{x}$$

$$\frac{3}{4} = \frac{80-y}{x}$$

$$3x = 80 - 4y \quad \text{—③}$$

Since along horizontal, $a_x = 0$, $V_x = 28\,m/s$

we have, $V_y^2 - u_y^2 = 2a_y y$ ($u_y = 0$)

$$V_y^2 = 2gy$$

$$\boxed{V_y = \sqrt{2gy}} \quad \text{—①}$$

$$x = V_x t = 28t$$

& $y = u_y t + \frac{1}{2} a_y t^2$ ($u_y = 0$)

$$y = \frac{1}{2} g t^2$$

$$\boxed{t = \sqrt{\frac{2y}{g}}} \Rightarrow x = 28\sqrt{\frac{2y}{g}} \quad \text{—②}$$

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$y = \frac{1}{2}gt^2$, $x = 28t$

$3(28t) = 80 - y(\frac{1}{2}gt^2)$

$42t = 40 - gt^2$

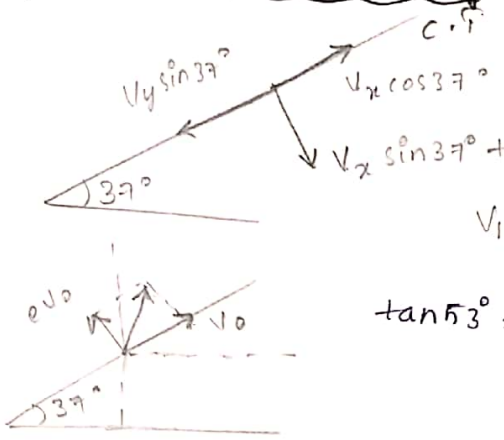
$gt^2 + 42t - 40 = 0$

$5t^2 + 21t - 20 = 0 \Rightarrow \boxed{t = 0.8\,sec}$

$V_y = u_y + a_y t$

$V_y = gt = 10 \times 0.8 = 8\ m/s$.

* resolution of velocity :



$V_x \sin 37° + V_y \cos 37° = V_\perp = 28 \times \frac{3}{5} + 8 \times \frac{4}{5} = \frac{84+32}{5} = \frac{116}{5}$.

$V_{||} = 17.6\ m/s$

$= 23.2\ m/s$

$\tan 53° = \frac{4}{3} = 1.33$. here $\tan\phi = \frac{eV_\perp}{V_{||}} = \frac{e(23.2)}{17.6}$

$\tan\phi = e(1.32)$

clearly $0 \le e \le 1$.

i.e, $\tan\phi < \tan 37°$

$\boxed{\phi < 53°}$

→ The particle hits the plane ⊥ler to A after rebound from the inclined plane it again hits at B. if e b/w particle and plane is $\frac{1}{\sqrt{2}}$. then find i) V = ____.

ii) AB = ____.

i) 1) $v_x = u_x + a_x t$.

$0 = V\cos 45° - g\cos 45°(t)$

$\boxed{t = \frac{V}{g}}$

2) $y = u_y t + \frac{1}{2}a_y t^2$

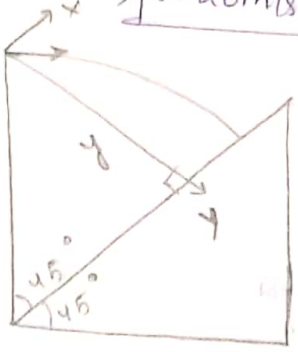
$y = (u\sin 45°)\left(\frac{V}{g}\right) + \frac{1}{2}g\cos 45°\frac{V^2}{g^2}$

$\frac{60}{\sqrt{2}} = \frac{V^2}{\sqrt{2}} \times \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{g}\right) + \frac{1}{2} \times \frac{1}{\sqrt{2}} \frac{V^2}{g}$

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$\frac{2V^2}{2g} = 60$ ii)

$\Rightarrow \boxed{V = 20 m/s}$



$\frac{2V^2}{2g} = 60 \Rightarrow \boxed{V = 20 m/s}$

+ Along X-axis, $y = 0$

$y = u_y t + \frac{1}{2} a_y t^2$

$0 = -eVt + \frac{1}{2} g\cos 45° t^2$

$t = \frac{2eV}{g\cos 45°} = \frac{2\sqrt{2}eV}{g} = \frac{2\sqrt{2} \times \frac{1}{\sqrt{2}} \times 20}{10} = 4 sec$

Along incline,

$x = u_x t + \frac{1}{2} a_x t^2$

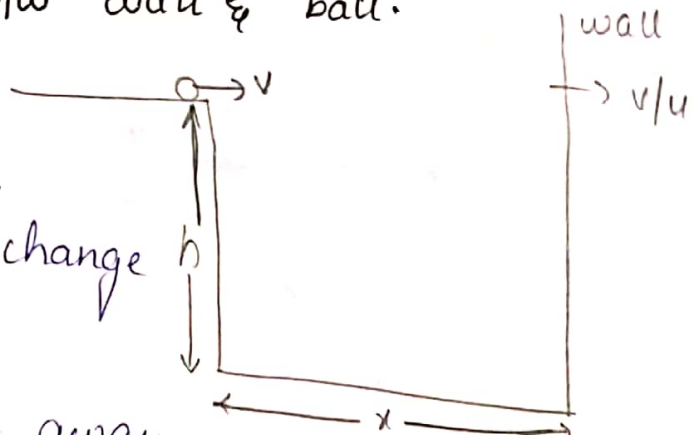
$x = \frac{1}{2}(-g\sin 45°) t^2$

$= -\frac{1}{2} \times 10 \times \frac{1}{\sqrt{2}} (4)^2 = -5 \times 4 \times \sqrt{2} \times 2 = -40\sqrt{2}$

A particle is thrown horizontally at height h towards a vertical wall as shown. if the particle returns to the point of projection after

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suffering 2 elastic collisions one with the wall, other with the ground. Find the total time of flight, intial seperation b/w wall & ball.

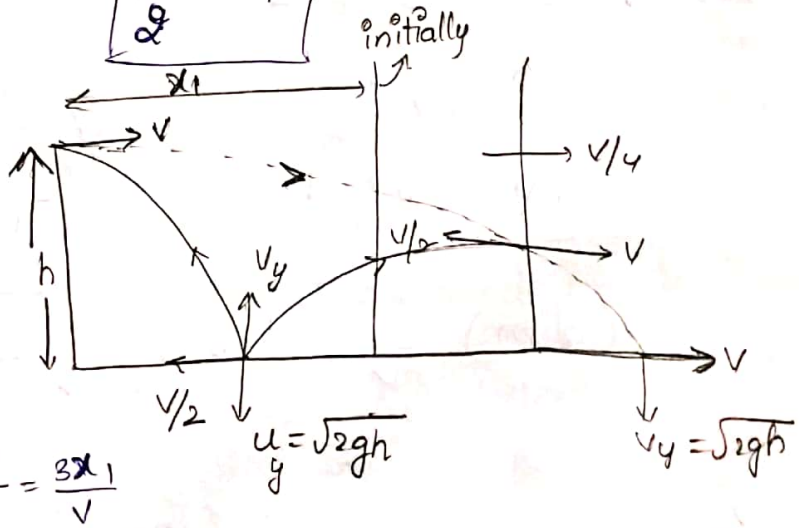


* key points:
* For massive wall, velocity doesnot change on collision.
* If wall is moving away, then 1st collision occurs on wall & then next collision on ground.
* If wall is moving towards particle, collision 1st occurs with ground and then with wall.

$$e = \frac{v_2 - v_1}{u_1 - u_2} \Rightarrow 1 = \frac{\frac{v}{4} - (-v_1)}{v - \frac{v}{4}}$$

$$\frac{3v}{4} = \frac{v}{4} + v_1$$

$$\boxed{\frac{v}{2} = v_1}$$



① Time of flight,

$$T = 2\sqrt{\frac{2h}{g}}$$

② Again, $T = t_1 + t_2$

$$T = \frac{x_1}{v} + \frac{x_1}{\frac{v}{2}} = \frac{3x_1}{v}$$

$$2\sqrt{\frac{2h}{g}} = \frac{3x_1}{v} \Rightarrow \boxed{x_1 = \frac{2v}{3}\sqrt{\frac{2h}{g}}}$$

from figure, $x + \frac{v}{4}t_1 = x_1 \Rightarrow x = x_1 - \frac{v}{4}t_1$

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$$x = \frac{2v}{3}\sqrt{\frac{2h}{g}} - \frac{v}{4}\sqrt{\frac{2h}{g}}$$

$$= v\sqrt{\frac{2h}{3}}\left(\frac{2}{3} - \frac{1}{4}\right) = \frac{8-3}{12}\, v\sqrt{\frac{2h}{g}} = \frac{5}{12}\, v\sqrt{\frac{4}{g}}$$

$$x = x_1 - \frac{x}{4} \times \frac{x_1}{x} = \frac{3x_1}{4}$$

$$x = \frac{2v}{3}\sqrt{\frac{2h}{g}} \times \frac{3}{4_2}$$

$$\boxed{x = \frac{v}{2}\sqrt{\frac{2h}{g}}}$$

→



① find max compression in the spring

② Max K.E of m_3.

③ Min K.E of m_2.

LCLM b/w ① & ②

$$mv = 2mv_2 \qquad 2m\frac{v}{2} = 3mv_c$$

$$\boxed{V_2 = v/2} \qquad \boxed{V_c = v/3}$$

$$\frac{1}{2}(2m)\frac{v^2}{4_2} = \frac{1}{2}kx_{max}^2 + \frac{1}{2}(3m)v_c^2$$

$$\frac{mv^2}{2} = kx_{max}^2 + 3mv_c^2$$

$$\frac{mv^2}{2} - 3m\frac{v_0^2}{93} = kx_{max}^2$$

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$$\frac{mv^2}{6} = kx_{max}^2$$

$$\boxed{x_{max} = \sqrt{\frac{m}{6k}}\, v}$$

Max K.E of m_3 is at it's natural length and m_2 has min K.E at this instant.

$$2m\left(\frac{v}{2}\right) = 2mv_3 + mv_4$$

$$v = 2v_3 + v_4 \quad -(1)$$

$$e = 1 = \frac{v_4 - v_3}{v/2} \Rightarrow \frac{v}{2} + v_3 = v_4 \quad -(2)$$

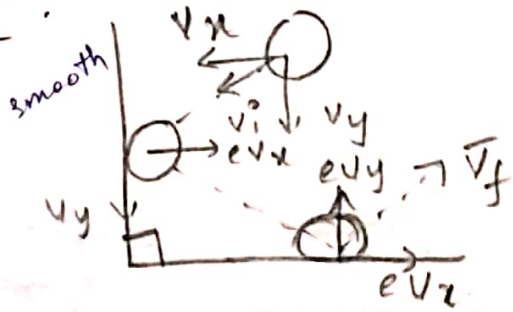
from (1) & (2)

$$\boxed{v_4 = \frac{2v}{3}}, \quad \boxed{v_3 = \frac{v}{6}}$$

$$(K.E)_3 = \frac{1}{2}m\left(\frac{v^2}{6}\right) = \frac{mv^2}{9}$$

$$(K.E)_2 = \frac{1}{2}(2m)\left(\frac{v^2}{36}\right) = \frac{mv^2}{36}$$

→ Smooth L-shaped obstacle placed on horizontal plane (coefficient of restitution is e). Find $\vec{v}_f$.



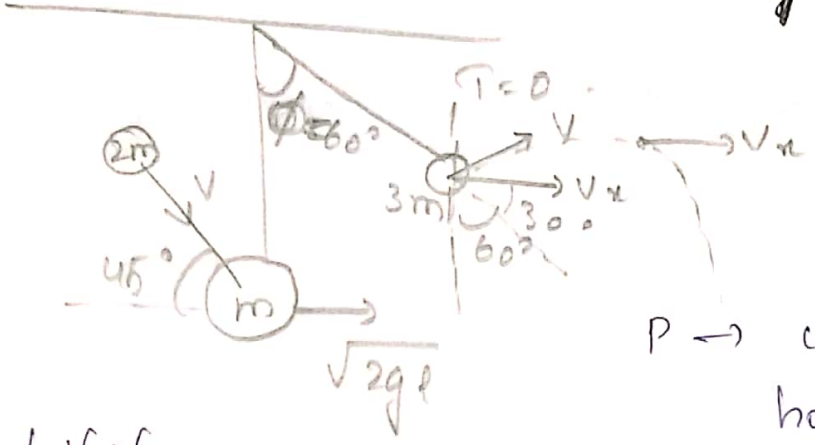
$$\vec{v}_f = ev_x(-\hat{i}) + ev_y(-\hat{j})$$

$$= -e(v_x\hat{i} + v_y\hat{j})$$

$$\boxed{\vec{v}_f = -e\,\vec{v}_i}$$

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→ A ball of mass m is suspended by a massless string to fixed point a ball of mass 2m strikes in the direction $\theta = 45°$ from horizontal and strikes to it if the initial velocity of 2m is $x\sqrt{ge}$ so that system deflects by $\pi/2$ and if string is cut at $\phi = 60°$ and the velocity at heighest point of trajectory is $\sqrt{\frac{gl}{y^2}}$ find $x+y$.



L.C.E,

$\frac{1}{2}(3m)(2gl) = \frac{1}{2}(3m)v^2 + 3mgl(1-\cos\phi)$

$3gl = \frac{3}{2}v^2 + 3gl\left(\frac{1}{2}\right)$

$\frac{3gl}{2} = \frac{3}{2}v^2$

$\boxed{v = \sqrt{gl}}$

P → conservation along horizontal direction

$2mv\cos 45° = 3m\sqrt{2gl}$

$v = 3\sqrt{gl}$

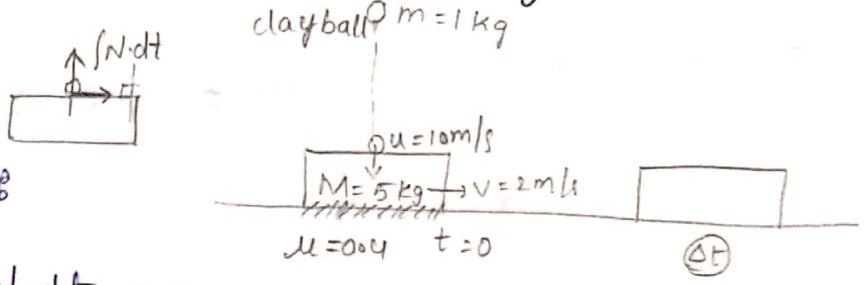
$\boxed{x = 3}$

$v_x = v\cos 60° = \frac{\sqrt{gl}}{2}$ $\Rightarrow \boxed{y = 2}$

$x + y = 5$

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→ A clay ball of mass 1kg is released on a block hitting it with a vertical velocity $u = 10 m/s$ at the instant of hit block has horizontal velocity $v = 2 m/s$ after an interval of Δt. another similar clay ball hits the block & the sys comes to rest immediately after the hit. Find Δt ($g = 10 m/s^2$).



on ball,

* vertical impulse:

$$mu - \int N \cdot dt = 0$$

$$\int N \cdot dt = mu \quad -(1)$$

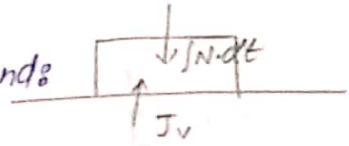
* Along horizontal,

$$0 + J_H' = m_1 v_1 \quad -(2)$$

v_1 = speed of ball after collision.

on block,

* vertical impulse on block due to ground:

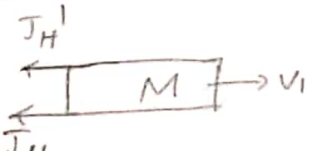


$$0 + \int N \cdot dt - J_V = 0$$

$$J_V = \int N dt = mu \quad -(3)$$

$$J_V = (1)(10) = 10 N\text{-}s$$

* Horizontal impulse on block:



$$MV - J_H = -J_H' = MV_1$$

↳ v_1 = speed of block just after impact

$$MV - J_H = mv_1 + MV_1$$

$$MV - J_H = (m+M)v_1 \quad -(4)$$

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- J_H = friction impulse on block due to ground

Again we have,

$$\boxed{J_H = \mu J_V} - (5)$$

$$\left[f = \mu N \Rightarrow \int f \cdot dt = \int \mu N \cdot dt \Rightarrow \int f \cdot dt = \mu \int N \cdot dt \right]$$

* Net impulse on block due to ground,

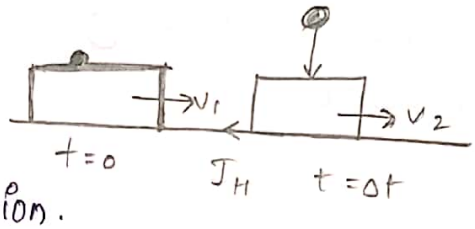
$$J = \sqrt{J_V^2 + J_H^2} = \sqrt{(10)^2 + (4)^2} = \sqrt{116}\ N\text{-}S.$$

$J_V = 10\ N\text{-}S$, $J_H = \mu J_V = 0.4 \times 10 = 4\ N\text{-}S$ & $(5)(2) - 4 = (1+5)v$

$$\boxed{v_1 = 1\ m/s}$$

Let (v_2) be the speed of block just before second hit by clay ball.

Here, J_V & J_H are not changing in each collision.



$$\therefore (M+m)v_2 - J_H = (M+2m)v_f \quad [v_f = 0]$$

$$(5+1)v_2 - 4 = 0$$

$$\boxed{v_2 = 2/3\ m/s}$$

Finally,

$$v_2 = v_1 + at$$

$$v_2 = v_1 - \mu g(\Delta t)$$

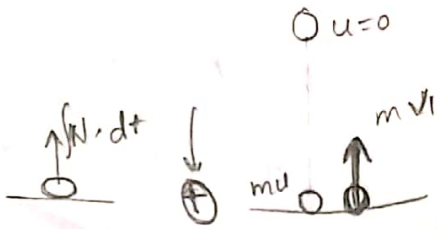
$$\frac{2}{3} = 1 - 0.4 \times 10 \times \Delta t$$

$$4\Delta t = \frac{1}{3}$$

$$\boxed{\Delta t = \frac{1}{12}\ sec}$$

*

$$\boxed{mu - \int N \cdot dt = -mv_1}$$



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Impulse due to Tension:

① $\int T \cdot dt$ = impulse due to tension, acting on the body which always acts "along the string connecting the bodies.

② Impulse due to tension ⊥er to string is "always zero".

③ Impulse = 0 ⇒

If impulse = 0

$\Delta P = 0$

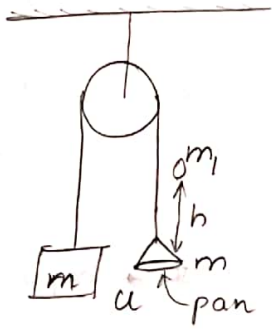
$\boxed{P = \text{const}}$ $\Rightarrow mv = \text{const}$

$\Rightarrow \boxed{v = \text{const}}$

④ string is inextensible.

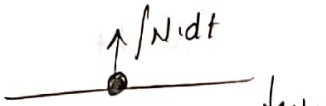
∴ velocity components/acceleration components of bodies connecting the string are same.

→ particle hits the point (pan) and sticks with it.



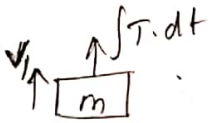
* Impulse eqns:

for a particle:



$$mu - \int N \cdot dt = mv_1 \quad -①$$

for a block:



$$0 + \int T \cdot dt = mv_1 \quad -②$$

for pan:



$$0 + \int N \cdot dt - \int T \cdot dt = mv_1 \quad -③$$

from eqns ①, ② & ③,

$m_1u - m_1v_1 - mv_1 = mv_1$

$m_1u = (2m + m_1)v_1$

$$\boxed{v_1 = \frac{m_1 u}{m_1 + 2m}}$$

— common velocity.

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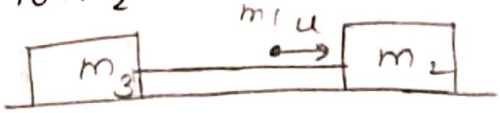
Impulse due to normal rxn,

$$\int N \cdot dt = m_1 u + m_1 v_1$$

Tension impulse,

$$\int T \cdot dt = m v_1 = \frac{m m_1 u}{2m + m_1}.$$

→ After collision, m_1 sticks to m_2.



Impulse eqns:

on m_1:-

$$m_1 u - \int N \cdot dt = m_1 V \quad -(1)$$

on m_2:-

$$0 + \int T \cdot dt = m_2 V \quad -(2)$$

on m_3:

$$0 + \int N \cdot dt - \int T \cdot dt = m_2 V \quad -(3)$$

(1) + (2) + (3)

$$m_1 u = m_1 V + m_2 V + m_3 V$$

$$\boxed{V = \frac{m_1 u}{m_1 + m_2 + m_3}}$$

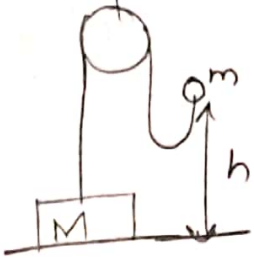
$$m_1 u - m_1 V = \int N \cdot dt$$

$$m_1 u - \frac{m_1^2 u}{m_1 + m_2 + m_3} = \int N \cdot dt$$

$$m_1 u \left[\frac{m_1 + m_2 + m_3 - m_1}{m_1 + m_2 + m_3} \right] = \int N \cdot dt$$

$$\boxed{\frac{m_1 (m_2 + m_3) u}{m_1 + m_2 + m_3} = \int N \cdot dt}$$

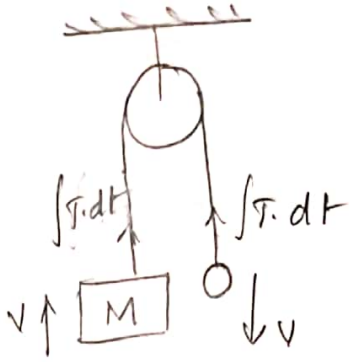
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Just string jerks

$u = \sqrt{2gh}$





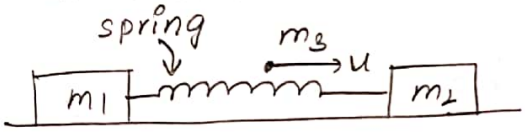
$$mu - \int T\,dt = mv \quad -①$$

$$0 + \int T\,dt = MV \quad -②$$

① + ② $\Rightarrow$ $mu = (m+M)V$

$$\boxed{V = \frac{mu}{m+M}}$$

→ m_3 sticks to m_2 after collision.



for m_3 :

$$m_3 u - \int N\,dt = m_3 V \quad -①$$

on m_2 : $0 + \int N\,dt - \int T\,dt = m_2 V$

$$\Rightarrow \int N\,dt = m_2 V \quad -②$$

on m_1 :

$$0 + \int T\,dt = m_1 V_1 \Rightarrow mV_1 = 0 \Rightarrow \boxed{V_1 = 0}$$

from eqn ① & ②

$$m_3 u = m_3 V + m_2 V$$

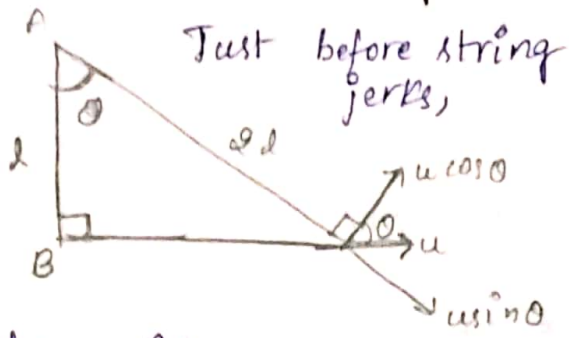
$$\boxed{V = \frac{m_3 u}{m_3 + m_2}}$$

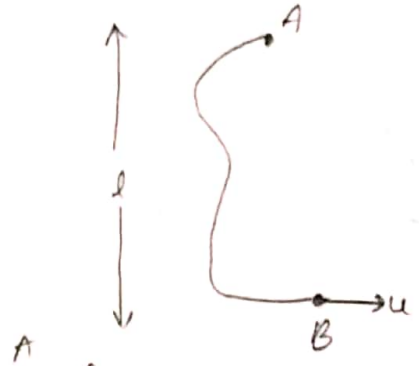
→ A, B are two identical particles having mass m connected by a string of length 2l. The arrangement is kept on the smooth horizontal table as shown. Just after string jerks. Find

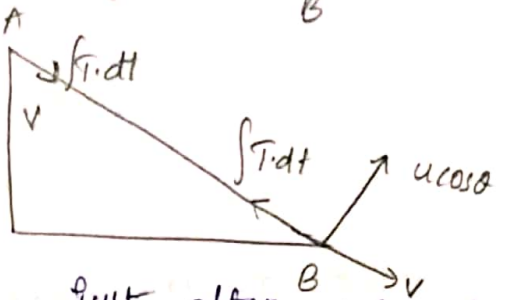
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① speed v_A & v_B ② Tension impulse in string.

③ Tension in string.







Just after string jerks.

Impulse eqn:

on A :-

$$0 + \int T\cdot dt = mv \quad -①$$

on B :-

$$mu\sin\theta - \int T\cdot dt = mv \quad -②$$

Adding eqns ① & ②,

$$mu\sin\theta = 2mv$$

$$\boxed{v = \frac{u}{2}\sin\theta}$$

③ $$\int T\cdot dt = mv$$

$$\int T\cdot dt = \frac{mu\sin\theta}{2}$$

* Finally,

① speed of A

$$v_A = v = \frac{u\sin\theta}{2}$$

② speed of B.

$$v_B = \sqrt{v^2 + (u\cos\theta)^2}$$

④ Tension in string:

We use centre of mass frame.

⇒ ⊥ler to string,

$$v_{cm} = \frac{mu\cos\theta}{2m} = \frac{u\cos\theta}{2}.$$

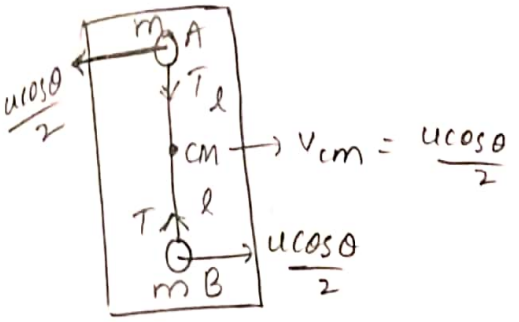
⇒ along the string,

$$v_{cm} = v.$$

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* $V_{A/cm} = V_A - V_{cm} = 0 - \frac{u\cos\theta}{2} = -\frac{u\cos\theta}{2}$.

* $V_{B/cm} = V_B - V_{cm} = u\cos\theta - \frac{u\cos\theta}{2} = \frac{u\cos\theta}{2}$.

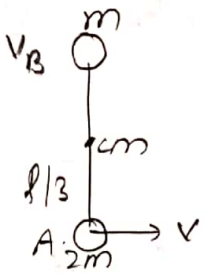


w.r.to COM, blocks are in circular motion.

$\therefore T = \frac{m\left(\frac{u\cos\theta}{2}\right)^2}{l} = \frac{mu^2\cos^2\theta}{4l}$

→ Find the tension in string.

$V_{cm} = \frac{2m(v)}{m+2m} = \frac{2v}{3}$



$V_{A/cm} = V_A - V_{cm}$

$= v - \frac{2v}{3} = \frac{v}{3}$

$V_{B/cm} = V_B - V_{cm}$

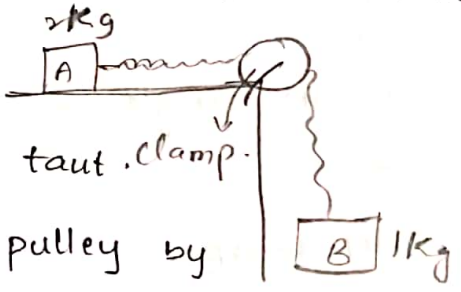
$x_{cm} = \frac{2m}{3m}(l) = l/3$

$T = \frac{mv^2}{r} = \frac{2m(v/3)^2}{l/3} = \frac{2mv^2/9}{l/3} = \frac{2mv^2}{3l}$

$$\boxed{T = \frac{2mv^2}{3l}}$$

→ System is released from rest and string becomes taut when B fall a distance 0.5m. Find

1) The common velocity (V_c) of two blocks just after string becomes taut.

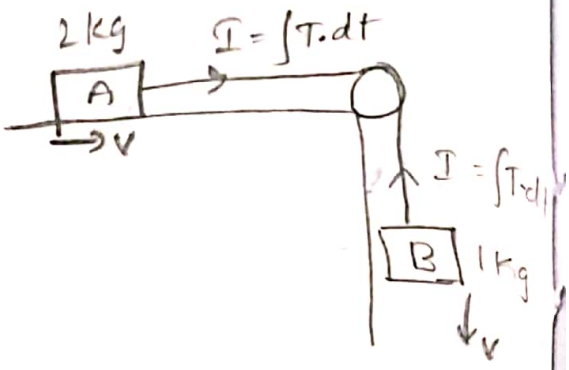


2) Magnitude of impulse on pulley by clamp during the small interval while string becomes taut.

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* just before string taut,

$$v_B = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 0.5}$$
$$= \sqrt{9.8} \text{ m/s}$$



Impulse eqn:

On B:

$$m_B v_B - I = m_B v \quad \text{—①}$$

on A:

$$I = m_A v \quad \text{—②}$$

$$m_B v_B = (m_A + m_B) \cdot v$$

$$v = \frac{1 \times \sqrt{9.8}}{3} = \frac{\sqrt{9.8}}{3}.$$

$$I = \int T \cdot dt = \frac{2}{3}\sqrt{9.8} \text{ N-S}$$

* Net impulse on pulley $= \sqrt{2} I = \frac{\sqrt{2}\sqrt{2 \times 9.8}}{3}$

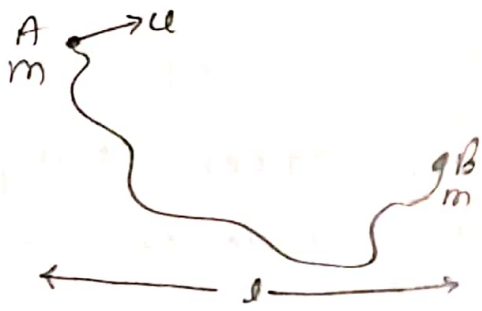
$$= \frac{2}{3}\sqrt{9.8} \text{ N-S}$$

→ String length = 2l. Arrangement is on smooth surface. A particle 'A' is projected accross the table with velocity 'u'. Find the speed with which the 2nd particle begins to move if the direction of 'u' is

① along BA

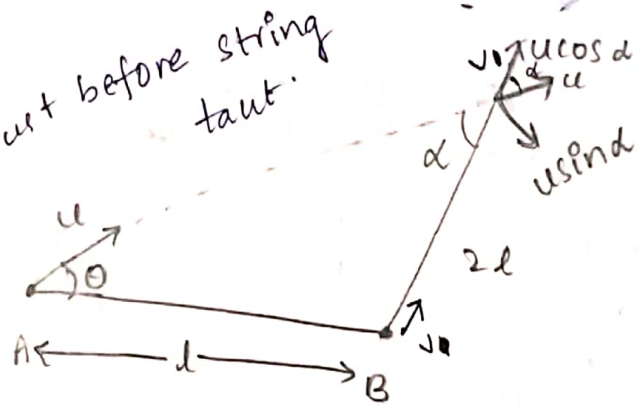
② at an angle of 120° with AB.

③ ⊥ler to AB.



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Just before string taut.

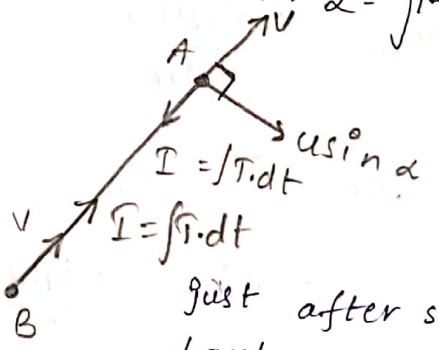


from sine rule,

$$\frac{2l}{\sin\theta} = \frac{l}{\sin\alpha}$$

$$\sin\alpha = \frac{\sin\theta}{2}.$$

$$\cos\alpha = \sqrt{1-\sin^2\alpha} = \sqrt{1-\frac{\sin^2\theta}{4}}$$



just after string taut.

Impulse Eqn:

On A:

$$mu\cos\alpha - \int T\,dt = mv \quad \text{—(1)}$$

On B:

$$0 + \int T\,dt = mv \quad \text{—(2)}$$

(1) + (2) $\quad mu\cos\alpha = 2mv$

$$\boxed{v = \frac{u\cos\alpha}{2}}$$

Speed of A –

$$v_A = \sqrt{v^2 + (u\sin\alpha)^2}$$

$$= \sqrt{\frac{u^2\cos^2\alpha}{4} + u^2\sin^2\alpha}$$

$$= u\sqrt{\cos^2\alpha + 4\sin^2\alpha}$$

$$v_A = u\sqrt{1+3\sin^2\alpha}$$

Speed of B –

$$v_A = v = \frac{u\cos\alpha}{2}$$

Impulse tension on string –

$$I = \int T\,dt = mv$$

$$\int T\,dt = \frac{mu\cos\alpha}{2}.$$

(1) $\theta = \pi$ rad $\Rightarrow \sin\alpha = \frac{\sin 180}{2} = 0 \Rightarrow \boxed{\alpha = 0}$

$$v_A = \frac{u}{2}\sqrt{1+3(0)} = \frac{u}{2}, \quad v_B = \frac{u}{2}\cos 0 = u/2.$$

$$I = \frac{mu}{2}\cos\theta = \frac{mu}{2}.$$

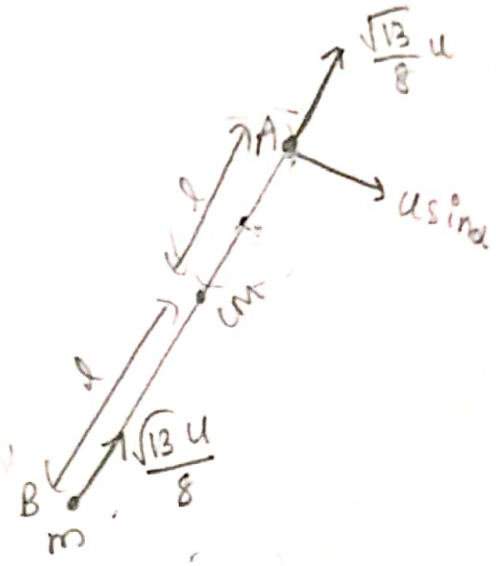
(2) $\theta = 120^\circ \Rightarrow \sin\alpha = \frac{\sin 120}{2} = \sqrt{3}/4, \quad \cos\alpha = \sqrt{1-\frac{3}{16}} = \frac{\sqrt{13}}{4}$

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$$V_A = \frac{u}{2}\sqrt{1+3\left(\frac{3}{16}\right)} = \frac{u}{2}\sqrt{\frac{25}{16}} = \frac{u}{2}\times\frac{5}{4} = \frac{5u}{8}.$$

$$V_B = \frac{u}{2}\times\frac{\sqrt{13}}{4} = \frac{\sqrt{13}\,u}{8}.$$

$$\int T\,dt = \frac{mu}{2}\cos\alpha = \frac{\sqrt{13}}{8}mu$$



* $V_{cm} = \frac{mu\sin\alpha}{2m} = \frac{u\sin\alpha}{2}$

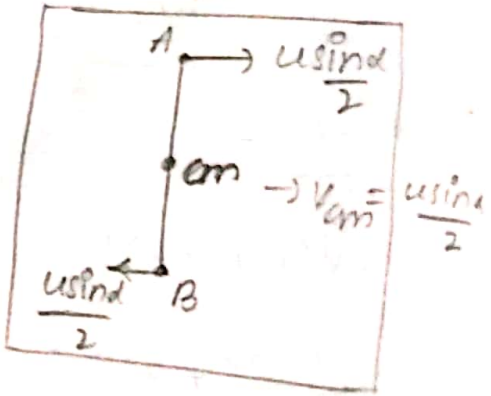
$$\overline{V_{A/cm}} = \overline{V_A} - \overline{V_{cm}}$$

$$= u\sin\alpha - \frac{u\sin\alpha}{2}$$

$$= \frac{u\sin\alpha}{2}.$$

$$\overline{V_{B/cm}} = \overline{V_B} - \overline{V_{cm}} = 0 - \frac{u\sin\alpha}{2} = -\frac{u\sin\alpha}{2}.$$

* Blocks are in circular motion about centre of mass.



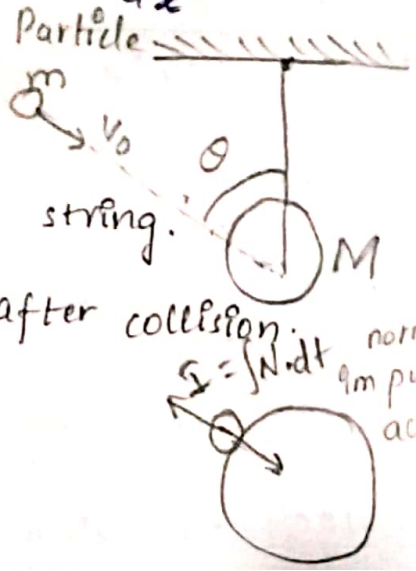
Tension in string,

$$T = \frac{m\left(\frac{u\sin\alpha}{2}\right)^2}{l} = \frac{mu^2\sin^2\alpha}{4l}$$

→ Particle comes to rest after collision. Find

① Impulse tension in the string.

② Velocity of bob just after collision.



At the time of collision,

Impulse eqn:

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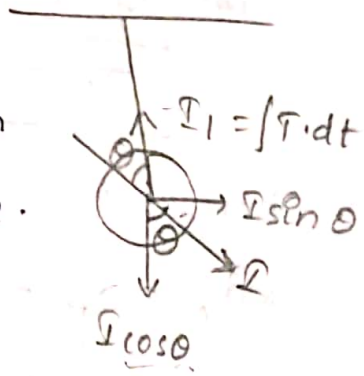
on ball :

$$mv_0 - I = 0$$

$$\boxed{I = mv_0}$$

on bob :

Since there is no change in momentum of bob along vertical.



$$I_1 = \int T\,dt = I\cos\theta$$

$$\boxed{I_1 = mv_0\cos\theta}$$

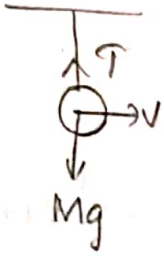
along horizontal,

$$I\sin\theta = MV$$

$$mv_0\sin\theta = MV$$

$$\boxed{V = \frac{mv_0\sin\theta}{M}}$$

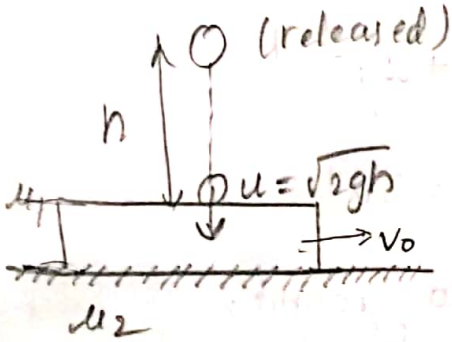
Tension in the string :



$$T - Mg = \frac{MV^2}{\ell}$$

$$\boxed{T = Mg + \frac{MV^2}{\ell}}$$

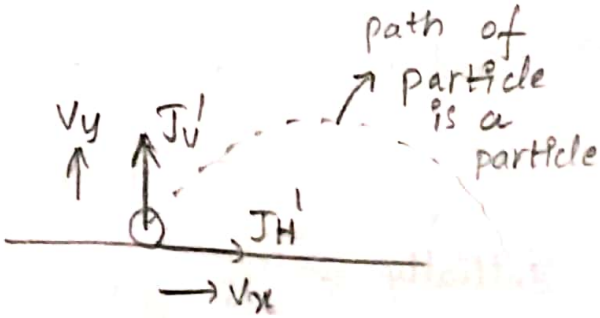
→



All surfaces are rough and particle bounce up from the block after hitting.

Impulse eqns :

On ball :



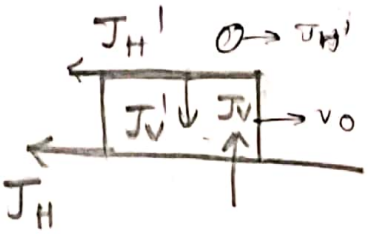
Along vertical,

$$mu - J_V' = -mv_y \quad -①$$

Along horizontal, $0 + J_H' = mv_x \quad -②$

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On block:



Since there is no change in momentum of block along vertical

$$\boxed{J_V = J_V'} \quad -(3)$$

along horizontal,

$$MV_0 - J_H' - J_H = MV \quad -(4)$$

Solving eqns (1), (2), (3) & (4). we get unknowns V_x, V_y, V & J_V

net impulse on the particle,

$$= \sqrt{(J_V')^2 + (J_H')^2}$$

$$= J_V' \sqrt{1+\mu_1^2}$$

Net impulse on block (by both particle & ground)

$$= J_H + J_H' = \mu_2 J_V + \mu_1 J_V'$$

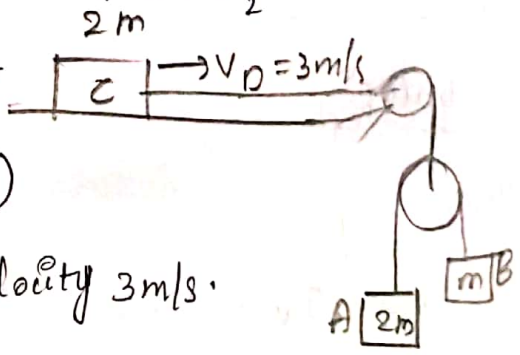
$$= (\mu_1+\mu_2) J_V \quad [\because J_V' = J_V]$$

Net impulse on block due to ground,

$$= \sqrt{J_V^2 + J_H^2} = J_V \sqrt{1+\mu_2^2}$$

→ Find V_A, V_B, V_C just after string is taut ($g = 10\ m/s^2$)

Initially (c) is given velocity 3m/s.



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string becomes taut when distance travelled by 'c' = distance fallen by pulley.

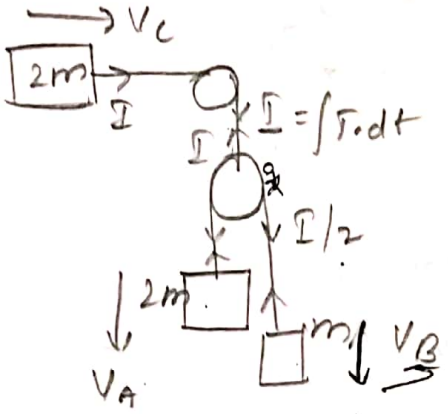
i.e., $V_0 (t) = \frac{1}{2} g t$

$$V_0 = \frac{1}{2} g t \Rightarrow t = \frac{2V_0}{g} = \frac{2(3)}{10} = 0.6 \text{ sec.}$$

Just before string taut,

$$V_A = V_B = gt = 10(0.6) = 6 \text{ m/s.}$$

just after string become taut,



Impulse eqns:

on C:

$$2mV_0 + \bar{I} = 2mV_C \quad -(1)$$

on A:

$$2mu_A - \frac{\bar{I}}{2} = 2mV_A \quad -(2)$$

on B:

$$mu_B - \frac{\bar{I}}{2} = mV_B \quad -(3)$$

Constraint eqn:

w.r.t pulley, $\bar{V}_{A/P} = -\bar{V}_{B/P}$

$$\bar{V}_A - \bar{V}_P = -(\bar{V}_B - \bar{V}_P)$$

$$\bar{V}_A + \bar{V}_B = 2V_P \quad -(4)$$

After solving,

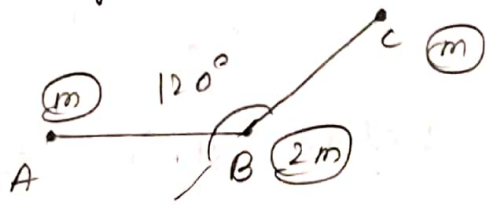
$V_A = \frac{36}{7}$ m/s

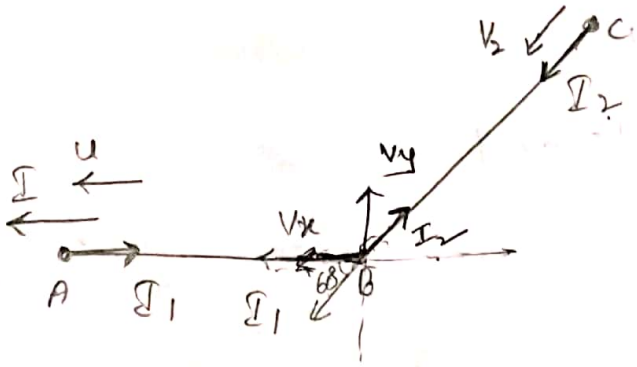
$V_B = \frac{30}{7}$ m/s

$V_C = \frac{33}{7}$ m/s; $\bar{I} = \frac{24}{7}$ mm N-s.

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→ Arrangement is on smooth horizontal surface. An Impulse '$\mathcal{I}$' is applied to point 'A' along BA so that it aquires velocity 'u'. Find Immediate velocity of 'B' & 'C'





constraint eqns -

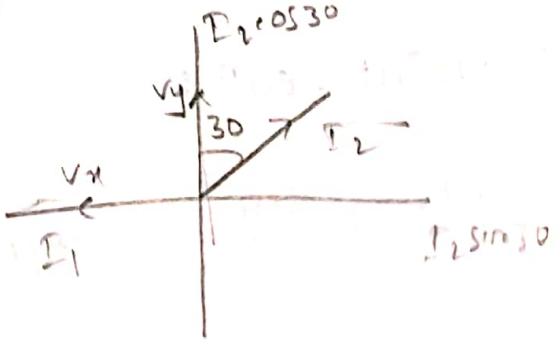
① $V_x = u$; $V_x \cos 60 - V_y \cos 30 = V_2$

Impulse eqn on A

$\mathcal{I} - I_1 = mu$ → ③

$I_2 = mv_2$ → ④

$\mathcal{I} - I_2 \sin 30 = 2mv_2$ → ⑤



unknowns

V_2 V_y V_x $\mathcal{I}$ I_2

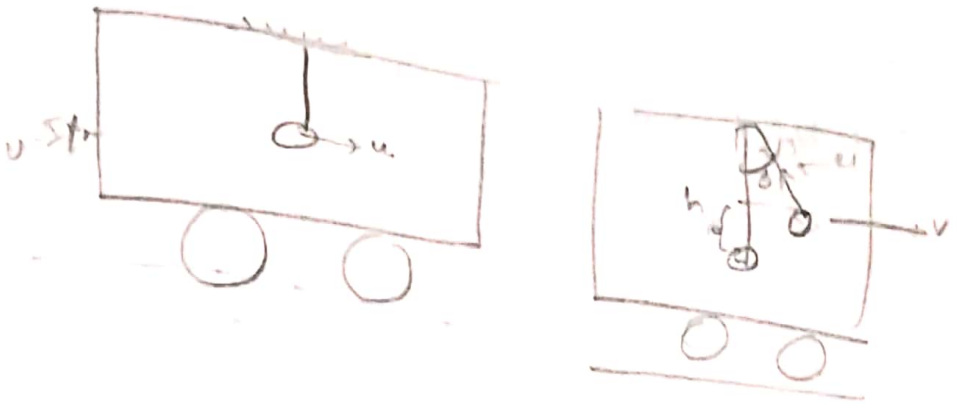
After solving

we get

$V_B = \frac{2\sqrt{31}}{11} u$

$V_C = \frac{4u}{11}$

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At max height relative to the cart, $mu = mV + MV \Rightarrow \boxed{V = \frac{mu}{m+M}}$

Tension in the string wrt cart frame,

Speed of bob is zero

$a_c = \frac{v^2}{R} = 0$

L.C.E,

$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + \frac{1}{2}MV^2 + mgh$

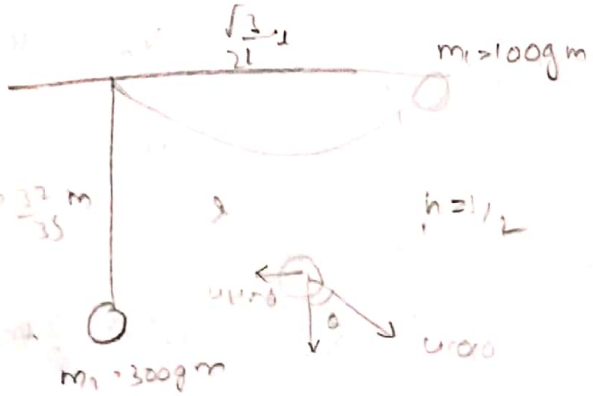
$a_{cart} = \frac{T\sin\theta}{M}$

Since $a_c = 0$

$T + f_{pseudo}\sin\theta = mg\cos\theta \Rightarrow T + \frac{mT\sin^2\theta}{M} = mg\cos\theta$

find ① velocity of m_1 just before collision with m_2 = 4 m/s

② If collision is elastic find max height raised by com of m_2



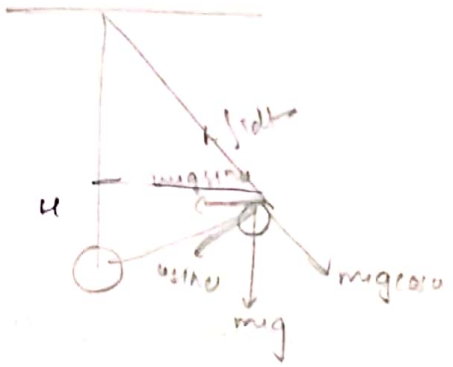
Impulse eqn

$mu\cos\theta - \int T\,dt = 0$ Impulsive tension

$\int T\,dt = mu\cos\theta$

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Just before string becomes taut

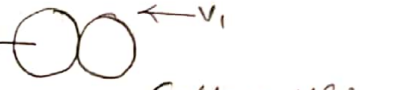


LCE blw B & C

$$mgh + \frac{1}{2}m_1(u\sin\theta)^2 - \frac{1}{2}m_1u_1^2$$

$$h = l(1-\cos\theta) = l(1-\cos 60^\circ) = \frac{l}{2}$$

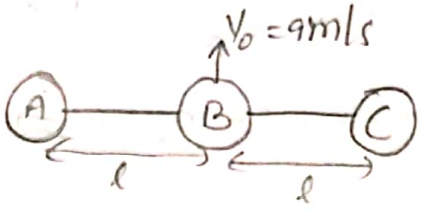
$$v_2 = \left(\frac{2m_1}{m_1+m_2}\right)u = \text{2 m/s.}$$

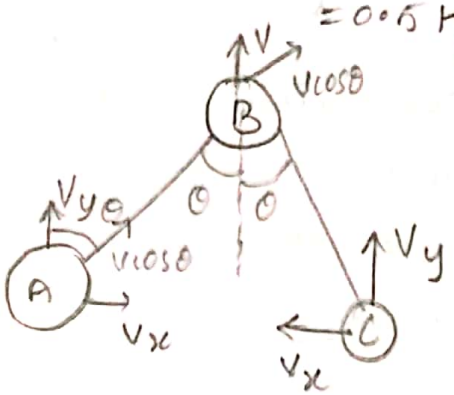


(after collision)

→ Find the speeds of A, B & C when the body A collides C.

$m_A = m_B = m_C = 0.5$ kg





Constraint eqn,

$v_y\cos\theta = v\cos\theta$

$$\boxed{v_y = v}$$

L.C.L.M, (along vertical)

$$mv_0 = mv + mv_y + mv_y$$

$$v_0 = v + 2v_y$$

$$v_0 = 3v_y \Rightarrow \boxed{v_y = \frac{v_0}{3}}$$

From L.C.E,

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv^2 + \frac{1}{2}m(v_x^2 + v_y^2) + \frac{1}{2}m(v_x^2 + v_y^2)$$

$$v_0^2 = v^2 + 2v_x^2 + 2v_y^2$$

$$v_0^2 = 2v_x^2 + 3v_y^2$$

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$$V_0^2 = 2V_x^2 + 3\left(\frac{V_0^2}{9}\right)$$

$$\frac{2V_0^2}{3} = 2V_x^2$$

$$\boxed{V_x = \frac{V_0}{\sqrt{3}}}$$

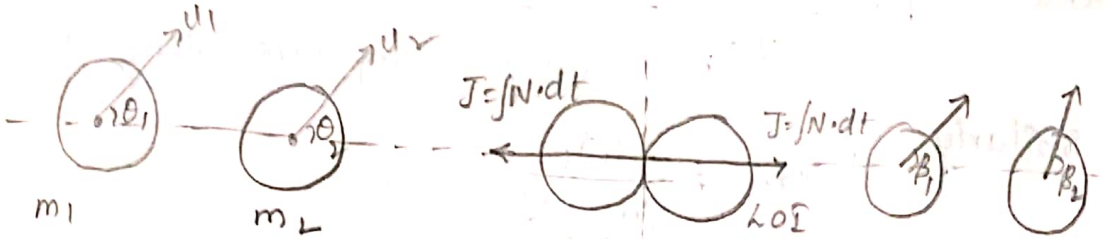
Speed of B_1, $V_B = V = \frac{V_0}{3} = 3\,m/s$

speed of A, Speed of $C = \sqrt{V_x^2 + V_y^2}$

$$= \sqrt{\frac{V_0^2}{3} + \frac{V_0^2}{9}}$$

$$= \frac{2V_0}{3} = \frac{2(9)}{3} = 6\,m/s.$$

Collision in 2D :



[Before collision] C.T [during collision] [after collision]

* L.C.L.M along x-axis,

$$m_1u_1\cos\theta_1 + m_2u_2\cos\theta_2 = m_1v_1\cos\beta_1 + m_2v_2\cos\beta_2 \quad \text{—(1)}$$

L.C.L.M along y-axis,

$$m_1u_1\sin\theta_1 + m_2u_2\sin\theta_2 = m_1v_1\sin\beta_1 + m_2v_2\sin\beta_2 \quad \text{—(2)}$$

* L.C.K.E,

$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \quad \text{—(3)}$$

* $e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{v_2\cos\beta_2 - v_1\cos\beta_1}{u_1\cos\theta_1 - u_2\cos\theta_2}$ —(4)

* L.C.L.M

* L.C.K.E (elastic collision)

* $e = \frac{v_2 - v_1}{u_1 - u_2}$ (along L.O.I)

* If colliding surfaces are smooth then velocity components of each body along C.T is same just before & after collision.

* Constraint eqn.

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* Along C.T,

$$u_1 \sin\theta_1 = v_1 \sin\beta_1 \quad \& \quad u_2 \sin\theta_2 = v_2 \sin\beta_2$$

Unknowns: $v_1, v_2, \beta_1, \beta_2$

After solving,

$$v_1 \cos\beta_1 = \left(\frac{m_1 - e m_2}{m_1+m_2}\right) u_1 \cos\theta_1 + \frac{m_2(1+e)}{m_1+m_2} u_2 \cos\theta_2 .$$

& $$v_2 \cos\beta_2 = \frac{m_1(1+e)}{m_1+m_2} u_1 \cos\theta_1 + \frac{m_2 - e m_1}{m_1+m_2} u_2 \cos\theta_2 .$$

* speed of body,

$$v_1 = \sqrt{(v_1 \sin\beta_1)^2 + (v_1 \cos\beta_1)^2}$$

$$v_2 = \sqrt{(v_2 \sin\beta_2)^2 + (v_2 \cos\beta_2)^2}$$

and $\tan\beta_1 = \frac{(v_1)_t}{(v_1)_n} = \frac{\text{velocity component along C.T}}{\text{velocity component along C.N}}$.

Similarly, $\tan\beta_2 = \frac{(v_1)_t}{(v_2)_n}$.

* Loss in K.E during collision = K.E$_i$ - K.E$_f$

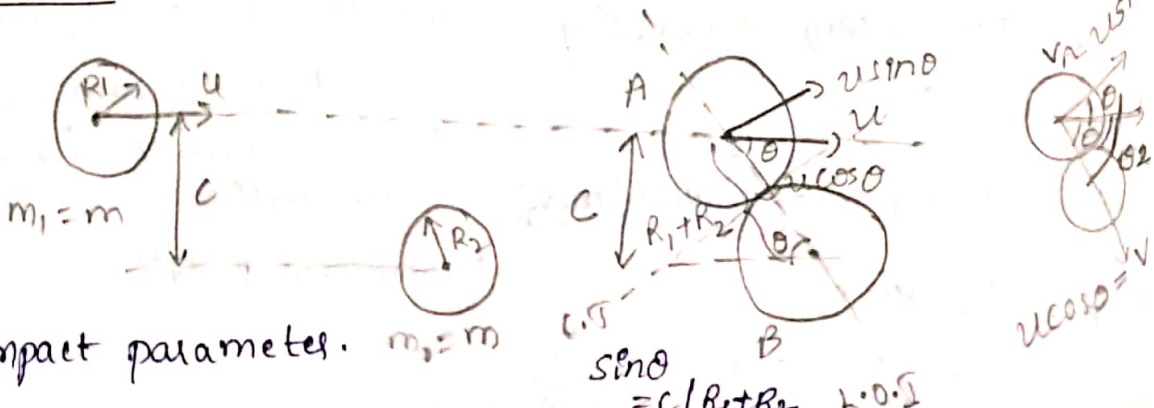
$$\Delta K = \frac{1}{2} \frac{m_1 m_2}{m_1+m_2} (1-e^2)(u_1 \cos\theta_1 - u_2 \cos\theta_2)$$

* Impulse ~~during~~ acting blw the bodies,

$$J = \frac{m_1 m_2}{m_1+m_2} (1+e)(u_1 \cos\theta_1 - u_2 \cos\theta_2) .$$

Applications:

①



C = Impact parameter.

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* If $C < R_1 + R_2$, then collision takes place.

θ_1 = scattering angle of m_1

θ_2 = scattering angle of m_2

$\theta_1 + \theta_2$ = diverging angle blw m_1 & m_2.

* If collision is elastic, velocities exchange during collision along L.O.I (for $m_1 = m_2$)

* L.C.L.M, along horizontal,

$$mu = mv_A \cos\theta_1 + mv_B \cos\theta_2$$

$$u = v_A \cos\theta_1 + v_B \cos\theta_2 \quad -(1)$$

* along vertical,

$$0 = mv_A \sin\theta_1 - mv_B \sin\theta_2$$

$$\Rightarrow 0 = v_A \sin\theta_1 - v_B \sin\theta_2 \quad -(2)$$

* L.C.E,

$$\frac{1}{2} mu^2 = \frac{1}{2} mv_A^2 + \frac{1}{2} mv_B^2$$

$$u^2 = v_A^2 + v_B^2 \quad -(3)$$

Squaring and adding eqns (1) & (2) —

$$u^2 = v_A^2 \{\cos^2\theta_1 + \cos^2\theta_2 + 2\cos\theta_1 \cos\theta_2\} + v_A^2 \sin^2\theta_1 + v_B \sin^2\theta_2 - 2v_A v_B \sin\theta_1 \sin\theta_2$$

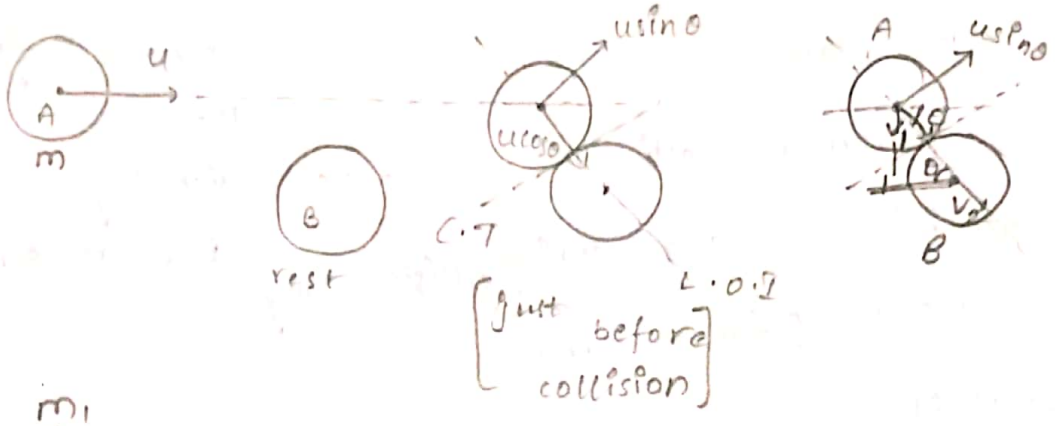
$$u^2 = v_A^2 + v_B^2 + 2v_A v_B \cos(\theta_1 + \theta_2)$$

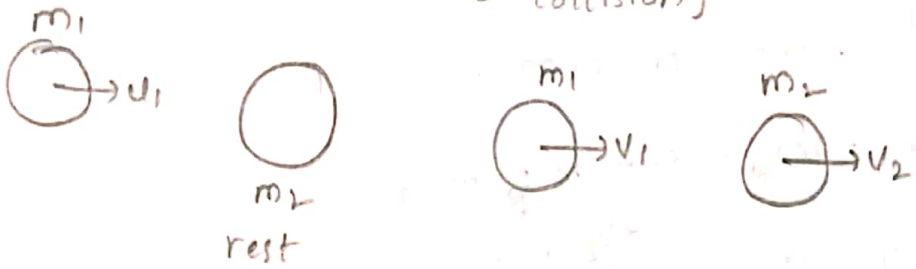
$$\cos(\theta_1 + \theta_2) = 0$$

$$\boxed{\theta_1 + \theta_2 = 90^\circ} \rightarrow \text{standard result}$$

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* If collision is Inelastic :





(1) $e = \frac{v_2 - v_1}{u\cos\theta}$ — (1)

(2) L.C.L.M, along horizontal,

$$mu = mv_1\cos\theta + m(u\sin\theta) + mv_2\cos\theta \quad — (2)$$

Solving (1) & (2) we get v_1 & v_2.

* Speed of B, $v_B = v_2$

Speed of A, $v_A = \sqrt{v_1^2 + u^2\sin^2\theta}$

* Impulse, $J = mv_2$.

* Loss in K.E,

$$= K.E_i - K.E_f$$

$$= \frac{1}{2}mu^2 - \frac{1}{2}mv_A^2 - \frac{1}{2}mv_B^2.$$

* If collision is perfectly Inelastic :

* L.C.L.M along L.O.I,

$$mu\cos\theta = mv + mv \Rightarrow \boxed{v = \frac{u\cos\theta}{2}}$$

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* speed of 'B', $v_B = \frac{u\cos\theta}{2}$

* speed of 'A', $v_A = \sqrt{(u\sin\theta)^2 + \left(\frac{u\cos\theta}{4}\right)^2}$

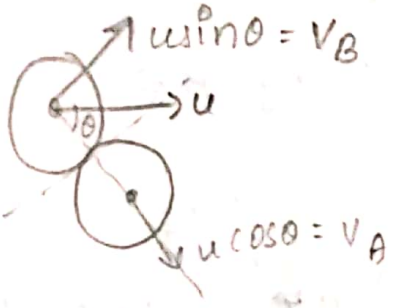
* Impulse, $J = mv_B = \frac{mu\cos\theta}{2}$

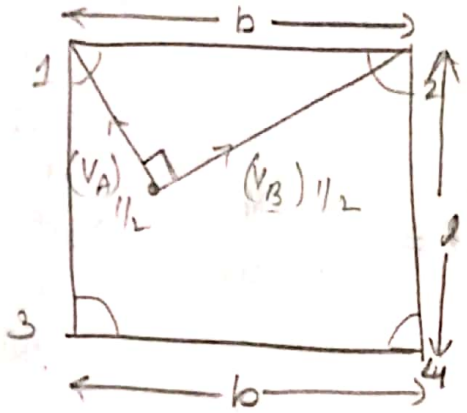
* Max loss in KE $= \frac{1}{2}mu^2 - \frac{1}{8}mv_A^2 - \frac{1}{2}mv_B^2$.

→ A billiard ball 'A' is somewhere at rest on the table. Another identical ball 'B' is moving with speed 2 m/s collides with ball 'A' elastically. After collision, both A & B falls simultaneously in pockets 1 & 2 in ½ sec. Find the area of the table.

$$\left(\frac{v_A}{2}\right)^2 + \left(\frac{v_B}{2}\right)^2 = b^2$$

$$v_A^2 + v_B^2 = 4b^2 \quad -①$$





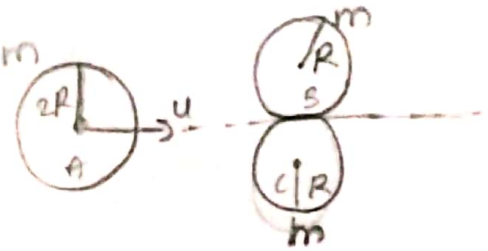
$$v_A^2 + v_B^2 = u^2 = 4$$

$$4b^2 = 4 \Rightarrow \boxed{b = 1}$$

& $\frac{l}{b} = 3 \Rightarrow \boxed{l = 3}$

Area $= lb = 3(1) = 3\ m^2$.

→



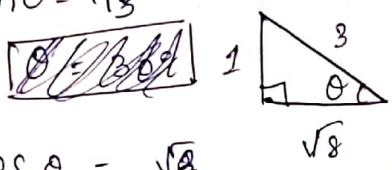
Disc A hits B & C symmetrically & comes to rest. Find $e = ?$.

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M-I:

$\sin\theta = \frac{R}{2R} = \frac{1}{2}$

$\sin\theta = 1/3$



$\cos\theta = \frac{\sqrt{8}}{3}$

From eqn ① & ②

$2J\cos\theta = mu$

$2mv\cos\theta = mu$

$v = \frac{u}{2\cos\theta}$

$\boxed{v = \frac{3u}{2\sqrt{8}}}$

$e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{\frac{3u}{2\sqrt{8}} - 0}{u\sqrt{8}/3}$

$e = \frac{9}{2(8)} = \frac{9}{16}$

$\boxed{e = \frac{9}{16}}$

Impulse eqn:

* For A:

$mu - 2J\cos\theta = 0$

$2J\cos\theta = mu$ — ①

* For B:

$J = mv$ — ②

Impulse on B/c:

$J = mv$

$= \frac{3mu}{2\sqrt{8}}$

Impulse on A:

$J' = mu = 2J\cos\theta$

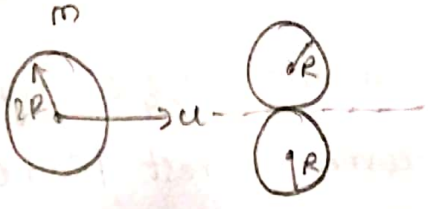
$\boxed{J' = mu}$

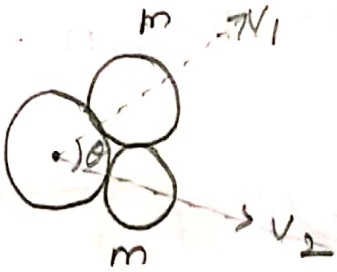
Loss of KE:

$= K.E_i - K.E_f$

$= \frac{1}{2}mu^2 - \left(\frac{1}{2}mv^2 + \frac{1}{2}mv^2\right)$

M-II:





* L.C.L.M: along x-axis,

$mu = mv_1\cos\theta + mv_2\cos\theta$ — ①

along y-axis, $0 = mv_1\sin\theta - mv_2\sin\theta$

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$\boxed{V_1 = V_2 = V}$ (say)

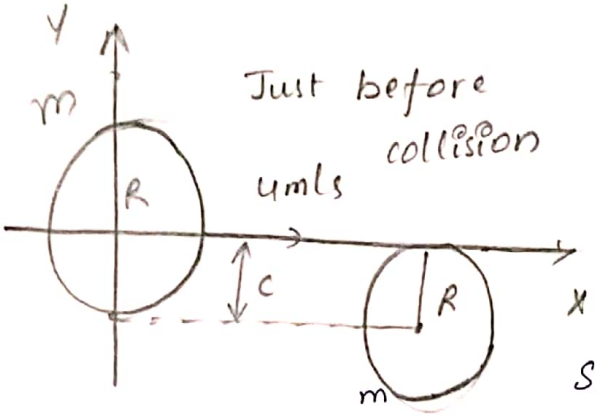
eqn ① becomes, $\boxed{u = 2V\cos\theta}$

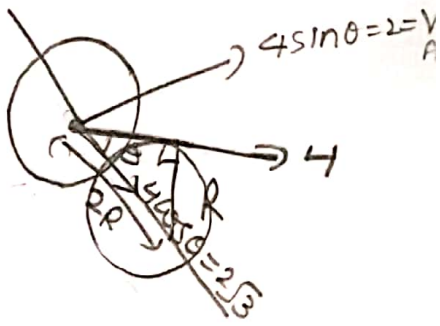
→ Two smooth balls A & B each of mass m & radius R having their centres ⇒ $C_A(0,0,R)$ & $C_B(2R,-R,R)$ Ball A is moving along +ve x-axis and collides with ball B. $V_A = 4$ m/s just before collision when $V_B = 0$. Collision blw balls is elastic. Find

① Velocity of balls A & B just after collision.

② Impulse on B due to 'A' during collision.

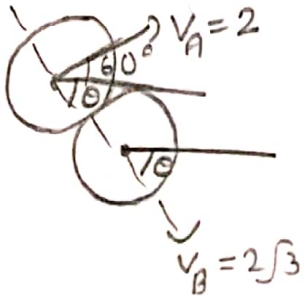
③ If e is changed to 1/2; keeping remaining parameters unchanged. find ~~(Speed)~~ Velocities of A & B just after collision.





$\sin\theta = \frac{R}{2R}$

$\boxed{\theta = 30^\circ}$



$\overline{V_A} = 2\cos 60^\circ \uparrow + 2\sin 60^\circ \rightarrow$

$\overline{V_A} = \uparrow + \sqrt{3}\rightarrow$

& $\overline{V_B} = 2\sqrt{3}\cos\theta\uparrow + 2\sqrt{3}\sin\theta(-\rightarrow)$

$= 3\uparrow - \sqrt{3}\rightarrow$

* Impulse on 'B' due to 'A'

$\overline{J} = m\overline{V_B} = m(3\uparrow - \sqrt{3}\rightarrow) = 3m\uparrow - \sqrt{3}m\rightarrow$

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